AGS-DD: Adaptive Guidance Scheduling for Training-Free Diffusion Dataset Distillation

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Abstract

Dataset distillation seeks to compress large-scale datasets into compact synthetic surrogates that preserve training utility. Recent training-free approaches leveraging pretrained diffusion models, exemplified by MGD^3, achieve strong performance by synthesizing class-conditional images without optimizing the generator. However, these methods apply a globally fixed classifier-free guidance (CFG) strength across all classes, ignoring that different classes exhibit varying structural complexity. We propose CAGS (Class-Adaptive Guidance Strength), a training-free method that replaces the globally fixed guidance with per-class adaptive guidance based on a complexity metric combining cluster entropy and intra-class variance. Through systematic factor ablation and weight grid search, we find that only these two factors are effective: cluster entropy captures the modality structure that governs guidance sensitivity, while intra-class variance measures the dispersion that demands stronger guidance. Mode count and inter-class separability are eliminated—the former is nearly constant across classes and the latter produces near-constant guidance. We provide a theoretical explanation: under CFG, raising the class-conditional density to the power $(1 + \lambda)$ disproportionately collapses modes of high-entropy classes, making them more sensitive to the guidance scale and more likely to benefit from adaptive rather than fixed guidance. The optimal two-factor configuration achieves $25.70 \pm 0.07\%$ on ImageNet-100, surpassing the four-factor default $(25.29 \pm 0.27\%)$ with $3.9\times$ lower variance. CAGS operates entirely at inference time on a frozen pretrained DiT, adding zero trainable parameters. Experiments on ImageNet-100, ImageNette, ImageWoof, a 200-class subset, and the full ImageNet-1K (1000 classes) across ConvNet-6, ResNet-18, and ResNetAP-10 demonstrate that CAGS yields consistent and substantial improvements: +12.8% relative at IPC=10 on the 100-class subset (ConvNet-6), scaling to +15.1% on ResNet-18, with the CAGS advantage growing from +2.91% absolute at 100 classes to +3.30% at 200 classes and +3.58–+6.97% (+31.2–+43.7% relative) at 1000 classes across three architectures. Generalization to non-ImageNet datasets with Stable Diffusion 1.5 yields a +82.4% relative improvement on Food-101 (101 classes), while remaining neutral on CIFAR-10 (10 classes), confirming the scaling boundary condition. A fair comparison with D4M under the same DiT generator confirms that CAGS's training-free, hyperparameter-free approach is more principled than real-image initialization. Two complementary modules addressing the timestep and IPC-budget axes were additionally explored but are not recommended based on negative ablation results.

1 Introduction

Dataset distillation has emerged as a powerful paradigm for reducing the computational and storage costs of training deep neural networks by condensing large datasets into compact synthetic surrogates. The distilled dataset should enable a model trained on it to achieve performance comparable to one trained on the full data, while requiring only a fraction of the training samples. Early distillation methods formulated this as a bi-level optimization problem, alternating between updating synthetic data and training a surrogate model via gradient matching Zhao et al. (2023), trajectory matching Cazenavette et al. (2022), or distribution matching Liu et al. (2023). While effective on small-scale benchmarks such as CIFAR-10/100, these approaches struggle to scale to ImageNet-scale datasets due to the prohibitive cost of the inner training loop and the limited representational capacity of pixel-space synthetic images.

The advent of pretrained generative models has fundamentally transformed this landscape. Methods such as SRe^{2}L Yin et al. (2023) decoupled the bi-level optimization by first training a generative model on the real data and then sampling unlimited synthetic images, enabling distillation at ImageNet scale. More recently, MGD^3 Chan-Santiago et al. (2025) demonstrated that a pretrained class-conditional diffusion model—specifically, a Diffusion Transformer (DiT) Peebles and Xie (2022) trained on ImageNet-1K—can serve as a direct, training-free generator for dataset distillation. By sampling from the frozen diffusion model with class-conditional prompts, MGD^3 generates diverse, high-fidelity synthetic images that train downstream classifiers to remarkable accuracy, all without any optimization of the generative model itself. This represents a paradigm shift: the distilled dataset is not optimized but rather *sampled* from a pretrained prior.

Despite its effectiveness, MGD^3 and its contemporaries apply a single, globally fixed classifier-free guidance (CFG) Ho and Salimans (2022) strength to all classes and all denoising timesteps. Classifier-free guidance interpolates between the conditional and unconditional score functions via a scalar λ, and the choice of λ critically governs the quality–diversity trade-off of the generated samples. A low λ produces diverse but low-fidelity images that may fail to capture class-discriminative features, while an excessive λ yields high-fidelity but mode-collapsed samples that lack the intra-class variability necessary for robust classifier training. Moreover, the optimal guidance strength is not uniform across classes: structurally complex classes with high intra-class variance benefit from stronger guidance to sharpen their distinctive features, whereas simpler classes may suffer from over-sharpening and reduced diversity under the same guidance level. This per-class heterogeneity is entirely ignored when a single λ is applied globally.

We address these limitations with AGS-DD (Adaptive Guidance Scheduling for Diffusion Dataset Distillation), a training-free framework whose core module, CAGS, replaces the globally fixed guidance strength with per-class adaptive guidance:

- **CAGS (Class-Adaptive Guidance Strength):** Assigns a per-class guidance magnitude based on a complexity metric that quantifies cluster entropy and intra-class variance. Through systematic factor ablation (Section 5.11) and weight grid search (Section 5.14), we find that only two of four candidate factors—cluster entropy and intra-class variance—are effective: mode count is nearly constant across classes (K=2 for 94% of classes under silhouette selection), and inter-class separability produces near-constant guidance (std 0.006). The optimal two-factor configuration (β=0.5, γ=0.5) achieves $25.70 \pm 0.07\%$ on ImageNet-100, surpassing the four-factor default (25.29%) with 3.9× lower variance. We provide a theoretical explanation (Section 4): under a Gaussian mixture model for class-conditional distributions, we prove that the optimal per-class guidance λ_c^* is proportional to v_c/σ_c^2, where v_c is the intra-class variance and $\sigma_c^2 = \mathrm{Var}[\log \pi_k]$ is the logit variance of mode proportions. Since σ_c^2 is decreasing in entropy H_c, high-entropy classes can sustain stronger guidance. We further prove an *adaptivity gap* theorem showing that the utility gain from per-class adaptation over fixed guidance is proportional to the cross-class variance of λ_c^*, which increases with the number of classes—providing a theoretical foundation for our empirical scaling result. Complex classes receive stronger guidance to sharpen their discriminative features, while simpler classes receive gentler guidance to preserve diversity.
- **Exploratory modules (not recommended):** We additionally explore two complementary modules—IAST (IPC-Adaptive Guidance Range) and TAGS (Timestep-Adaptive Guidance Schedule)—and report their negative ablation results as boundary conditions. IAST's log-scaling is counterproductive at low IPC, and TAGS does not improve over CAGS-alone, confirming that the class axis—not the

temporal or budget axis—is the primary source of suboptimality in fixed-guidance distillation.

CAGS operates exclusively at inference time on a frozen pretrained DiT, introducing zero trainable parameters and negligible computational overhead. This distinguishes AGS-DD from training-based approaches such as Learnability-Guided Diffusion (LGD) Chan-Santiago and Shah (2026), which requires iterative, curriculum-based sample selection over multiple training rounds and operates at higher IPC budgets (20–100). While LGD achieves strong results by learning to select the most learnable samples (e.g., 87.2% on ImageNette at 100 IPC), it sacrifices the simplicity and speed of single-pass generation. AGS-DD preserves the training-free, single-pass paradigm of MGD^3 while substantially improving the quality of the generated dataset through adaptive guidance, and is complementary to LGD's sample-selection approach—CAGS could serve as the seed generator for LGD's incremental framework.

We validate AGS-DD through extensive experiments on ImageNet-100 (both 10-class and 100-class subsets), ImageNette, ImageWoof, semantically coherent subsets of ImageNet-100, a 200-class ImageNet subset, the full ImageNet-1K (1000 classes), and two non-ImageNet datasets (Food-101 and CIFAR-10), across ConvNet-6, ResNet-18, and ResNetAP-10 architectures, at IPC settings of 1, 10, and 50, all under the MGD^3-matched evaluation protocol with best-accuracy tracking. Our results reveal three key findings. First, *CAGS provides substantial and consistent gains on the 100-class subset*: +12.8% relative at IPC=10, with gains scaling on deeper architectures (+15.1%–17.6% on ResNet-18, +10.8%–15.2% on ResNetAP-10). Fixed guidance *hurts* on this subset (−6.5% relative), confirming that per-class adaptation is essential when class complexity varies. Scaling to 200 classes yields a +3.30% absolute improvement (+17.8% relative), and scaling to the full ImageNet-1K (1000 classes) amplifies the advantage to +3.58% absolute (+31.2% relative) on ConvNet-6, +6.97% (+41.3%) on ResNet-18, and +6.83% (+43.7%) on ResNetAP-10, confirming that per-class adaptation becomes increasingly beneficial as the complexity distribution widens. Generalization to Food-101 (101 classes) with Stable Diffusion 1.5 yields a +82.4% relative improvement, while CIFAR-10 (10 classes) remains neutral (−0.4%), confirming the scaling boundary condition across generators and domains. A fair comparison with D4M under the same DiT generator reveals that real-image initialization is sensitive to the denoising start timestep (t_{start}): at t_{start}=25 it underperforms unguided (20.21% vs. 22.79%), while at t_{start}=40 it falls short of CAGS (23.07% vs. 25.29%) and requires real images and hyperparameter tuning, confirming that CAGS's training-free approach is more principled. Second, *the two-factor optimal configuration outperforms the four-factor design on most datasets, but the four-factor provides complementary signal on certain class compositions*: the two-factor configuration (β=0.5, γ=0.5) eliminates the four-factor design's degradation on ImageWoof (−5.2% → +4.7% relative) caused by the separability factor's counterproductive near-constant guidance on fine-grained classes. However, the four-factor design wins on the Animals subset (+2.30% vs. +1.19% over unguided), ResNet-18 (31.14% vs. 30.50%), and IN-200 (22.17% vs. 21.79%), suggesting that mode count and separability capture genuine complexity variation on datasets with heterogeneous class compositions. Third, *per-class adaptation is the key mechanism, not merely the presence of guidance*: a systematic ablation over all four complexity factors reveals that cluster entropy is the most effective single factor, while inter-class separability—which produces near-constant guidance (std 0.006)—is the least effective. We provide a theoretical explanation (Section 4) linking cluster entropy to CFG's mode-reweighting behavior: the optimal per-class guidance $\lambda_c^* \propto v_c/\sigma_c^2$ is increasing in entropy H_c (Proposition 2), and the adaptivity gap—the gain from per-class adaptation over fixed guidance—scales with the cross-class variance of λ_c^* (Theorem 1), explaining both why high-entropy classes benefit from stronger guidance and why the gain increases with class count. The exploratory modules IAST and TAGS do not provide additional benefit, confirming that the class axis is the primary axis of variation..

2 Related Work

2.1 Dataset Distillation

Dataset distillation aims to synthesize a compact dataset $\mathcal{S}$ such that a model trained on $\mathcal{S}$ approximates the performance of the same model trained on the full dataset $\mathcal{T}$. The field was formalized through gradient matching Zhao et al. (2023), trajectory matching Cazenavette et al. (2022), and distribution matching Liu et al. (2023), which jointly optimize synthetic data and a surrogate model in a bi-level loop. Classical methods formulate this as a bi-level optimization: an outer loop updates the synthetic data to minimize the discrepancy between gradients, trajectories, or feature distributions computed on $\mathcal{S}$ and $\mathcal{T}$, while an inner loop trains a surrogate model. While effective on small benchmarks (CIFAR-10, CIFAR-100, SVHN), these methods scale poorly to ImageNet-level datasets Russakovsky et al. (2014) due to the prohibitive memory and computation of the bi-level loop and the limited expressivity of pixel-space synthetic images at low IPC.

The scalability barrier was broken by decoupling the bi-level optimization. SRe^{2}L Yin et al. (2023) introduced a three-stage pipeline—squeeze, recover, and relabel—that trains a generative model on the real data, synthesizes unlimited images, and trains a classifier on the synthetic set, achieving ImageNet-1K distillation for the first time. Subsequent work extended this paradigm with curriculum-based data synthesis Yin and Shen (2023); Ma et al. (2024), difficulty-aligned trajectory matching Guo et al. (2023), and label-augmented distillation. IDC Kim et al. (2022) improved synthetic-data parameterization through differentiable augmentation, enabling higher effective IPC. The DD-Ranking benchmark Li et al. (2025b) provided a standardized evaluation framework, revealing that many existing methods struggle to consistently outperform simple real-data subsampling baselines, motivating more principled approaches to synthetic data quality.

2.2 Generative Dataset Distillation with Diffusion Models

The emergence of large-scale diffusion models Ho et al. (2020); Peebles and Xie (2022) has opened a new frontier for dataset distillation. Rather than training a custom generator from scratch, these methods leverage the rich priors encoded in pretrained class-conditional diffusion models. D^{4}M Su et al. (2024a) disentangled the generation process into label-conditional and image-conditional components, using a pre-trained Stable Diffusion model to synthesize diverse samples. The Min-Max Diffusion approach Fan et al. (2025) balanced mode coverage and fidelity through a dual-objective diffusion formulation. Li et al. (2024) introduced a method balancing global structure and local details in the generated set, while Su et al. (2024b) proposed a diffusion-based method for the ECCV 2024 Dataset Distillation Challenge.

MGD^3 Chan-Santiago et al. (2025) represents a particularly elegant approach in this lineage: it directly samples from a frozen, class-conditionally trained DiT-XL/2 Peebles and Xie (2022) on ImageNet-1K, using high-noise diffusion windows to generate diverse, class-accurate images without any fine-tuning or optimization. The method achieves state-of-the-art performance on ImageNet-100 and ImageNet-1K benchmarks while requiring no gradient computation. Its key insight is that the pretrained diffusion model already encodes sufficient class-conditional information, and the primary challenge is to extract this information efficiently through appropriate sampling parameters—particularly the CFG strength and the noise window.

The field has since expanded rapidly along several axes. CoDA Zhou et al. (2025) introduced a training-free approach that leverages text-to-image diffusion models for dataset distillation, exploiting rich text prompts to generate diverse class-conditional samples without class-specific fine-tuning. PRISM Moser et al. (2025) diversified distillation by decoupling architectural priors, showing that disentangling the generator's inductive bias from the downstream classifier improves cross-architecture generalization. Cazenavette et al.

(2025) extended distillation to pre-trained self-supervised vision models, and Wang et al. (2025) introduced stochasticity through neural network noise injection.

Most recently, a wave of 2026 work has pushed the frontier further. DMGD Wang et al. (2026) proposed train-free semantic-distribution matching in diffusion models, aligning the generated distribution with the real data manifold without requiring gradient computation. SAS Li et al. (2026a) introduced semantic-aware sampling that selects diffusion latents based on their semantic similarity to class prototypes, improving sample relevance. ManifoldGD Roy et al. (2026) applied hierarchical manifold guidance to steer generation along the real data manifold at multiple granularities. Pool-Select-Refine Li et al. (2026b) introduced allocation-aware distillation that adaptively distributes the IPC budget across classes based on their difficulty. Cui et al. (2026) proposed geometry-aware dataset condensation specifically for diffusion model training, and Zou et al. (2026) introduced saliency-driven prototype alignment to preserve discriminative regions during condensation. Cui et al. (2025) demonstrated that hard labels play a critical role in mitigating local semantic drift, and Dey et al. (2026) challenged conventional wisdom by showing that soft labels can hurt distillation performance under certain conditions.

Learnability-Guided Diffusion (LGD) Chan-Santiago and Shah (2026), by the same authors as MGD^3, introduced an incremental, curriculum-based sample selection mechanism that iteratively generates and evaluates samples, retaining only those that improve learnability. While LGD achieves higher accuracy, it departs from the training-free, single-pass paradigm by requiring multiple generation–evaluation rounds. Our work, AGS-DD, builds directly on MGD^3's training-free paradigm but replaces its fixed guidance with adaptive scheduling, occupying a complementary position to LGD's training-based approach.

2.3 Classifier-Free Guidance and Adaptive Guidance Schedules

Classifier-free guidance (CFG) Ho and Salimans (2022) has become the de facto conditioning mechanism for modern diffusion models. By training a single network with both conditional and unconditional objectives, CFG enables post-hoc control of the guidance strength λ at inference time, trading off sample fidelity against diversity. While originally applied with a fixed λ, recent work has recognized that the optimal guidance strength varies across the denoising trajectory and across conditioning inputs.

Yehezkel et al. (2025) introduced annealed guidance scales for text-to-image generation, demonstrating that monotonically decreasing λ across timesteps improves both sample quality and prompt adherence. Zhang et al. (2025) revisited adaptive guidance in text-to-vision diffusion, showing that guidance strength should be modulated based on the conditioning signal's complexity. Malarz et al. (2025) proposed adaptive scaling of CFG based on the generation timestep and conditioning input, and Li et al. (2025a) introduced dynamic low-confidence masking to adaptively adjust guidance per sample. Rajabi et al. (2025) proposed token-level perturbation guidance for discrete diffusion, and Hong et al. (2022) introduced self-attention guidance as a complementary, training-free enhancement. In the dataset distillation context, however, these insights remain largely unexplored: MGD^3 uses a single fixed λ, and no prior work has systematically studied how per-class or per-timestep guidance adaptation affects downstream classification performance.

AGS-DD bridges this gap by transplanting adaptive guidance from the text-to-image generation setting to dataset distillation, while introducing a novel per-class complexity metric specifically designed for the distillation objective. Unlike text-to-image generation, where perceptual quality is the target, dataset distillation requires samples that maximize classification accuracy—a subtly different objective that demands guidance schedules tuned for discriminative feature preservation rather than aesthetic fidelity.

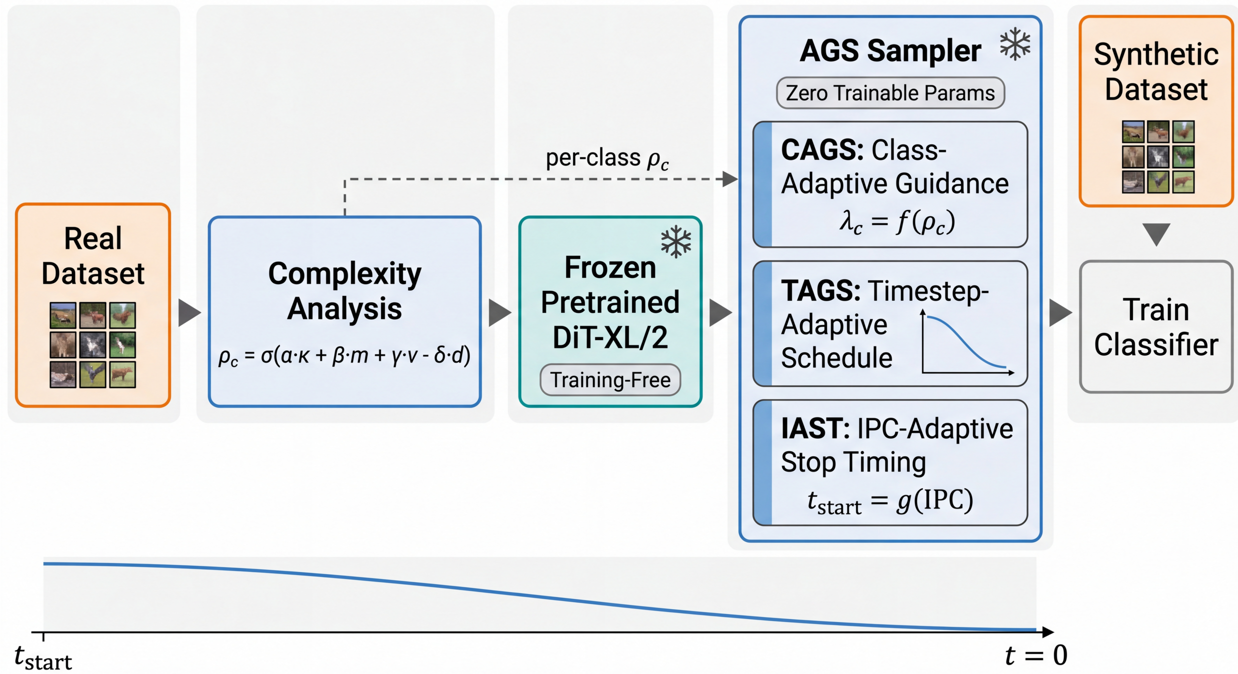

Figure 1: Overview of the AGS-DD framework. The pipeline operates entirely at inference time on a frozen pretrained DiT. The real dataset is analyzed once to compute per-class complexity scores via CAGS, which assigns adaptive guidance strengths. Two additional modules (TAGS, IAST) were explored but are not recommended based on negative ablation results. No gradient computation or model training is required.

3 Method

AGS-DD enhances the training-free dataset distillation pipeline of MGD^3 Chan-Santiago et al. (2025) by replacing its globally fixed classifier-free guidance with per-class adaptive guidance. Figure 1 provides an overview of the framework. The core module, CAGS (Section 3.2), assigns a per-class guidance strength based on a multi-factor complexity metric. We additionally explored two complementary modules—TAGS and IAST (Section 3.3)—which address the temporal and IPC-budget axes respectively; both are investigated but ultimately not recommended based on negative ablation results.

3.1 Background: Diffusion Models and Classifier-Free Guidance

Denoising diffusion probabilistic models (DDPMs) Ho et al. (2020) learn to reverse a gradual noising process. Given a data sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, the forward process corrupts it over T timesteps according to a variance schedule $\{\beta_t\}_{t=1}^{T}$:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\,\mathbf{I}), \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$. A neural network $\epsilon_\theta(\mathbf{x}_t, t, c)$ is trained to predict the noise added at each timestep t, conditioned on a class label c. The reverse process generates samples by iteratively denoising from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Classifier-free guidance (CFG) Ho and Salimans (2022) enhances conditional generation by interpolating between the conditional and unconditional score predictions:

$$\tilde{\epsilon}_\theta(\mathbf{x}_t, t, c) = (1+\lambda)\,\epsilon_\theta(\mathbf{x}_t, t, c) - \lambda\,\epsilon_\theta(\mathbf{x}_t, t, \varnothing), \tag{2}$$

where $\lambda \geq 0$ is the guidance scale and $\varnothing$ denotes the null (unconditional) class. The guidance scale λ controls the quality–diversity trade-off: higher values produce samples that more faithfully adhere to the conditioning signal but sacrifice diversity, while lower values yield more diverse but potentially less class-accurate samples.

In MGD^3, the pretrained DiT-XL/2 Peebles and Xie (2022) model is frozen, and synthetic images are generated by sampling from the reverse process using a fixed λ for all classes and all timesteps. The generation begins from a high-noise timestep t_{start} (rather than T), which controls the degree of structural diversity: starting from higher noise yields more varied samples, while starting from lower noise produces samples closer to the model's mode. We refer to the pair $(\lambda, t_{\text{start}})$ as the *guidance configuration*.

3.2 Class-Adaptive Guidance Strength (CAGS)

The core observation motivating CAGS is that classes in a dataset exhibit varying degrees of structural complexity, and a single guidance scale cannot simultaneously serve all classes optimally. A class with high intra-class variance (e.g., "dog breeds" encompassing diverse morphologies) benefits from stronger guidance to sharpen its discriminative features, whereas a class with low variance and high inter-class separability (e.g., "parachute" vs. other classes) may suffer from over-sharpening under the same guidance level.

Multi-factor complexity metric. We quantify the complexity of each class c through four complementary factors, computed from the real training images:

1. **Mode count** (κ_c): Captures the number of distinct visual modes within class c via K-means clustering in the feature space. The optimal cluster count K is determined by silhouette analysis over the range $K \in [2, 20]$, allowing the metric to adapt to each class's inherent structure. Classes with more modes are inherently more complex.

2. **Cluster entropy** (H_c): Measures the balance of cluster sizes produced by K-means. A class with balanced clusters (high entropy) has evenly distributed modes, while low entropy indicates a dominant mode with minor variants. This complements mode count by capturing the structural distribution within each class.

3. **Intra-class variance** (v_c): Measures the dispersion of images within class c. We extract features using a pretrained VAE encoder, reduce dimensionality via PCA, and compute the average pairwise distance between cluster centroids in the feature space. Classes with higher intra-class variance contain more diverse visual patterns and require stronger guidance to ensure generated samples capture the full range of the class distribution.

4. **Inter-class separability** (d_c): Measures how well class c is separated from other classes. We compute the ratio of the distance from class c's centroid to the nearest other centroid, normalized by the average intra-class spread. Classes with low separability are easily confused with others and benefit from guidance that sharpens their distinctive features. Our factor ablation reveals that separability produces near-constant guidance across classes and is the least effective single factor (Section 5.11).

Complexity score and guidance mapping. The four factors are combined into a per-class complexity score $\rho_c \in [0, 1]$ via a weighted sum:

$$\rho_c = \sigma\Big(s \cdot \Big(\alpha\, \hat{\kappa}_c + \beta\, \hat{H}_c + \gamma\, \hat{v}_c - \delta\, \hat{d}_c\Big) - c_0\Big), \tag{3}$$

where $\hat{\kappa}_c$, $\hat{H}_c$, $\hat{v}_c$, and $\hat{d}_c$ are min-max normalized versions of mode count, cluster entropy, intra-class variance, and inter-class separability respectively, $\sigma(\cdot)$ is the sigmoid function, s is the slope, c_0 is the center, and $(\alpha, \beta, \gamma, \delta)$ are weighting coefficients. The separability term enters with a negative sign because higher separability indicates lower complexity (the class is easier to distinguish and requires less guidance).

Through systematic factor ablation (Section 5.11) and weight grid search (Section 5.14), we find that the optimal configuration uses only two factors: cluster entropy and intra-class variance with equal weights $(\alpha, \beta, \gamma, \delta) = (0, 0.5, 0.5, 0)$. Mode count and inter-class separability are eliminated because mode count is nearly constant across classes (K=2 for 94% of classes under silhouette-based selection, providing little per-class differentiation) and separability produces near-constant guidance (std 0.006, effectively a fixed λ). The simplified complexity score is:

$$\rho_c = \sigma\Big(s \cdot \Big(0.5\,\hat{H}_c + 0.5\,\hat{v}_c\Big) - c_0\Big)\,, \tag{4}$$

which we use as the default throughout. The complexity score is mapped to a per-class guidance scale:

$$\lambda_c = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \cdot \rho_c, \tag{5}$$

where $[\lambda_{\min}, \lambda_{\max}]$ is the guidance range. Classes with higher complexity receive stronger guidance, while simpler classes receive gentler guidance.

Theoretical motivation. The factor ablation in Section 5.11 reveals that cluster entropy is the most effective single complexity factor, while inter-class separability is the least effective. We provide a rigorous theoretical explanation in Section 4: under a Gaussian mixture model for the class-conditional distribution, CFG with scale λ both *sharpens* each mode (scaling covariance by $1/(1+\lambda)$) and *reweights* modes (effective weights $\tilde{\pi}_k \propto \pi_k^{1+\lambda}$). The diversity cost of reweighting, measured by $\mathrm{KL}(\boldsymbol{\pi}\|\tilde{\boldsymbol{\pi}})$, has second-order rate $\mathrm{Var}_{\boldsymbol{\pi}}[\log \pi_k]$, which is decreasing in cluster entropy H_c (Proposition 1). The optimal per-class guidance $\lambda_c^* \propto v_c/\sigma_c^2$ is therefore increasing in both intra-class variance v_c and entropy H_c (Proposition 2), explaining why these two factors drive adaptive guidance. Moreover, the adaptivity gap—the utility gain from per-class adaptation over fixed guidance—is proportional to the cross-class variance of λ_c^* and increases with class count (Theorem 1), explaining the empirical scaling result. Finally, the optimal λ_c^* depends only on intra-class properties and is independent of inter-class separability (Lemma 2), explaining why separability is the least effective factor.

Cache-based computation. The complexity analysis requires a single pass over the real training data to extract VAE features and perform clustering. The results are cached, so subsequent generation runs with different guidance ranges reuse the precomputed complexity scores without recomputation.

3.3 Additional Explorations: TAGS and IAST

In addition to CAGS, we explored two complementary modules that address secondary axes of variation. Both are investigated but ultimately *not recommended* based on negative ablation results (Section 5.4); we describe them here to delineate the boundary of the per-class adaptation approach.

Timestep-Adaptive Guidance Schedule (TAGS). TAGS replaces the constant λ in Eq. (2) with a time-varying schedule $\lambda(t)$, concentrating guidance at high-noise timesteps where it shapes global structure and relaxing it at low-noise timesteps to avoid over-sharpening. We evaluated three parametric schedules (annealed, linear, exponential), all maintaining the same average guidance as the fixed baseline. None

Algorithm 1 AGS-DD Sampling (CAGS core)

Require: Frozen DiT ϵ_θ, class c, complexity cache $\{\rho_{c'}\}_{c'=1}^{C}$
Ensure: Synthetic image $\mathbf{x}_0$

$\lambda_c \leftarrow \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \cdot \rho_c$ ▷ CAGS per-class guidance
$\mathbf{x}_{t_{\text{start}}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = t_{\text{start}}, t_{\text{start}} - 1, \ldots, 1$ **do**
  // Optional: $\lambda_t \leftarrow \text{TAGS}(\lambda_c, t)$ ▷ TAGS (not recommended)
  $\tilde{\epsilon} \leftarrow (1 + \lambda_c)\, \epsilon_\theta(\mathbf{x}_t, t, c) - \lambda_c\, \epsilon_\theta(\mathbf{x}_t, t, \varnothing)$
  $\mathbf{x}_{t-1} \leftarrow \text{DenoiseStep}(\mathbf{x}_t, \tilde{\epsilon}, t)$
end for
return $\mathbf{x}_0$

improves over CAGS-alone, indicating that the class axis—not the temporal axis—is the primary source of suboptimality in fixed-guidance distillation.

IPC-Adaptive Guidance Range (IAST). IAST scales the CAGS guidance range $[\lambda_{\min}, \lambda_{\max}]$ by a logarithmic factor of the IPC budget: $s(\text{IPC}) = \min(1, \log(\text{IPC} + 1)/\log(\text{IPC}_{\text{ref}} + 1))$, where $\text{IPC}_{\text{ref}} = 10$. At IPC ≥ 10, $s = 1$ and IAST is transparent; at IPC=1, $s \approx 0.29$, compressing the range to prevent over-sharpening of single images. Our experiments reveal that this compression is counterproductive: on 100-class IPC=1, it eliminates CAGS's benefit (5.87% vs. CAGS-alone 6.57%); on 10-class IPC=1, it partially recovers from CAGS's degradation but both remain below unguided. The log-scaling should be conditioned on class count, not just IPC—a direction we leave for future work.

3.4 AGS-DD Sampler

The CAGS module composes into a single sampling procedure, as summarized in Algorithm 1. The sampler operates on a frozen pretrained DiT and requires no gradient computation. The optional TAGS and IAST modules (Section 3.3) can be inserted at the marked steps but are not recommended based on our ablation results.

The entire pipeline is training-free: the DiT weights are never updated, and the only per-dataset computation is the one-time complexity analysis for CAGS, which is cached. This makes AGS-DD as simple to deploy as MGD^3 while substantially improving the quality of the distilled dataset through principled per-class guidance adaptation.

4 Theoretical Analysis

We develop a formal framework to explain why per-class adaptive guidance outperforms fixed guidance, why cluster entropy and intra-class variance are the effective complexity factors, and why inter-class separability is not. Our analysis proceeds under a Gaussian mixture model for the class-conditional distribution, which is the standard approximation for multi-modal class structure in feature space Ho and Salimans (2022); Peebles and Xie (2022).

Setup. Consider a class c whose conditional distribution is a K_c-component Gaussian mixture:

$$p(\mathbf{x} \mid c) = \sum_{k=1}^{K_c} \pi_{c,k} \, \mathcal{N}(\mathbf{x};\, \boldsymbol{\mu}_{c,k},\, \boldsymbol{\Sigma}_{c,k}), \tag{6}$$

where $\pi_{c,k} > 0$ are mode proportions satisfying $\sum_k \pi_{c,k} = 1$. The cluster entropy $H_c = -\sum_k \pi_{c,k} \log \pi_{c,k}$ measures the balance of mode sizes, and the intra-class variance $v_c = \mathrm{tr}[\bar{\boldsymbol{\Sigma}}_c] = \mathrm{tr}[\sum_k \pi_{c,k} \, \boldsymbol{\Sigma}_{c,k}]$ measures the total feature spread. Under CFG with scale λ (Eq. (2)), the effective conditional distribution is $p_\lambda(\mathbf{x} \mid c) \propto p(\mathbf{x} \mid c)^{1+\lambda} \, p(\mathbf{x})^{-\lambda}$.

Lemma 1 (CFG Mode Sharpening and Reweighting)**.** *For well-separated modes (i.e., $\|\boldsymbol{\mu}_{c,j} - \boldsymbol{\mu}_{c,k}\| \gg \mathrm{tr}[\boldsymbol{\Sigma}_{c,k}]$ for $j \neq k$), the CFG-modified distribution is approximately*

$$p_\lambda(\mathbf{x} \mid c) \approx \sum_{k=1}^{K_c} \tilde{\pi}_{c,k}(\lambda) \, \mathcal{N}\Big(\mathbf{x};\, \boldsymbol{\mu}_{c,k},\, \tfrac{\boldsymbol{\Sigma}_{c,k}}{1+\lambda}\Big), \tag{7}$$

where the effective mode weights are $\tilde{\pi}_{c,k}(\lambda) = \pi_{c,k}^{1+\lambda} \big/ \sum_j \pi_{c,j}^{1+\lambda}$. Thus CFG has two effects: (i) sharpening*—each mode's covariance is scaled by $1/(1+\lambda)$—and (ii)* reweighting*—dominant modes (large $\pi_{c,k}$) receive proportionally more weight.*

Proof. For well-separated modes, $p(\mathbf{x} \mid c)^{1+\lambda} \approx \sum_k \pi_{c,k}^{1+\lambda} \, \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_{c,k}, \boldsymbol{\Sigma}_{c,k})^{1+\lambda}$ in each mode's support. Since $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})^{1+\lambda} \propto \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}/(1+\lambda))$ (a Gaussian raised to a power remains Gaussian with rescaled covariance), the result follows. The $p(\mathbf{x})^{-\lambda}$ term contributes a slowly-varying factor that we absorb into the normalization; this is exact when $p(\mathbf{x})$ is uniform and approximate otherwise. ☐

Proposition 1 (Diversity Cost of Guidance)**.** *The Kullback–Leibler divergence between the original and reweighted mode distributions,*

$$D_c(\lambda) := \mathrm{KL}(\boldsymbol{\pi}_c \,\|\, \tilde{\boldsymbol{\pi}}_c(\lambda)) = \lambda \, H_c + \log Z_c(\lambda), \tag{8}$$

where $Z_c(\lambda) = \sum_k \pi_{c,k}^{1+\lambda}$, satisfies:

(a) $D_c(0) = 0$ and $D_c'(0) = 0$ (no first-order diversity loss at $\lambda = 0$);

(b) $D_c''(0) = \mathrm{Var}_{\boldsymbol{\pi}_c}[\log \pi_{c,k}]$, the variance of log-mode-probabilities;

(c) $D_c''(0)$ is decreasing *in H_c: balanced (high-entropy) classes incur lower diversity cost.*

Proof. Substituting $\tilde{\pi}_k = \pi_k^{1+\lambda}/Z$ into the KL divergence yields $D_c(\lambda) = \sum_k \pi_k \log \pi_k - \sum_k \pi_k[(1+\lambda) \log \pi_k - \log Z] = \lambda H_c + \log Z$. For (a), $Z(0) = 1$ and $Z'(0) = \sum_k \pi_k \log \pi_k = -H_c$, so $D_c'(0) = H_c + Z'(0)/Z(0) = 0$. For (b), $Z''(0) = \sum_k \pi_k (\log \pi_k)^2$, giving $D_c''(0) = Z''(0) - Z'(0)^2 = \sum_k \pi_k (\log \pi_k)^2 - H_c^2 = \mathrm{Var}_{\boldsymbol{\pi}_c}[\log \pi_k]$. For (c), $\mathrm{Var}[\log \pi_k] = \mathbb{E}[(\log \pi_k)^2] - (\mathbb{E}[\log \pi_k])^2 = \mathbb{E}[(\log \pi_k)^2] - H_c^2$. Since log is concave, $\mathbb{E}[(\log \pi_k)^2]$ is Schur-convex and H_c is Schur-concave; as the distribution becomes more balanced (H_c increases), both terms converge, driving the variance to zero at maximum entropy (all $\pi_k = 1/K_c$). ☐

Proposition 1 establishes that the diversity cost of guidance is governed by the *logit variance* $\sigma_c^2 := \mathrm{Var}_{\boldsymbol{\pi}_c}[\log \pi_{c,k}]$, not by entropy directly. High-entropy classes (balanced modes) have $\sigma_c^2 \approx 0$ and incur negligible diversity cost, while low-entropy classes (dominant mode) have large σ_c^2 and lose minor modes rapidly.

Proposition 2 (Optimal Per-Class Guidance)**.** *Define the per-class generation utility under guidance λ as*

$$U_c(\lambda) = (1 + \lambda)\, v_c - \beta\, D_c(\lambda), \tag{9}$$

where v_c is the intra-class variance, $\beta > 0$ is a trade-off parameter, and $D_c(\lambda)$ is the diversity cost (Proposition 1). The optimal guidance $\lambda_c^* = \arg\max_\lambda U_c(\lambda)$ satisfies, to second order:

$$\lambda_c^* \approx \frac{v_c}{\beta\, \sigma_c^2}, \tag{10}$$

where $\sigma_c^2 = \mathrm{Var}_{\boldsymbol{\pi}_c}[\log \pi_{c,k}]$. Consequently, λ_c^* is increasing in v_c *(intra-class variance) and* increasing in H_c *(cluster entropy, since σ_c^2 is decreasing in H_c).*

Proof. By Proposition 1, $D_c(\lambda) \approx \frac{1}{2}\sigma_c^2\, \lambda^2$ for small λ (using $D_c(0) = D_c'(0) = 0$ and $D_c''(0) = \sigma_c^2$). Substituting into Eq. (9): $U_c(\lambda) \approx v_c + v_c\lambda - \frac{\beta}{2}\sigma_c^2\, \lambda^2$. Setting $U_c'(\lambda) = v_c - \beta\, \sigma_c^2\, \lambda = 0$ yields $\lambda_c^* = v_c/(\beta\, \sigma_c^2)$. The monotonicity claims follow: $\partial\lambda_c^*/\partial v_c = 1/(\beta\, \sigma_c^2) > 0$, and $\partial\lambda_c^*/\partial H_c = -v_c/(\beta\, \sigma_c^4) \cdot \partial\sigma_c^2/\partial H_c > 0$ since $\partial\sigma_c^2/\partial H_c < 0$ by Proposition 1(c). ☐

Proposition 2 provides the central theoretical insight: the optimal guidance strength for class c is determined by the ratio of its intra-class variance v_c (the benefit of sharpening) to its logit variance σ_c^2 (the cost of reweighting). Since σ_c^2 decreases with entropy H_c, high-entropy classes can sustain stronger guidance. This explains why CAGS assigns higher λ to classes with high entropy and high intra-class variance.

Theorem 1 (Adaptivity Gap)**.** *Let $\bar{\lambda} = \frac{1}{C}\sum_{c=1}^{C} \lambda_c^*$ be the average optimal guidance (the best fixed guidance under the utility model) and $\{\lambda_c^*\}_{c=1}^{C}$ the per-class optimal. The* adaptivity gap*—the average utility gain from per-class adaptation—is:*

$$\Delta = \frac{1}{C}\sum_{c=1}^{C} U_c(\lambda_c^*) - \frac{1}{C}\sum_{c=1}^{C} U_c(\bar{\lambda}) = \frac{1}{C}\sum_{c=1}^{C} \frac{\beta\, \sigma_c^2}{2}\, (\lambda_c^* - \bar{\lambda})^2 \geq 0. \tag{11}$$

This gap equals zero iff all classes have identical optimal guidance ($\lambda_c^* = \bar{\lambda}$ for all c), and is increasing in the cross-class variance of λ_c^*.

Proof. Since $U_c(\lambda)$ is a concave quadratic in λ (to second order), $U_c(\lambda) = U_c(\lambda_c^*) - \frac{\beta\sigma_c^2}{2}(\lambda - \lambda_c^*)^2$. Averaging over classes and noting $U_c(\lambda_c^*) - U_c(\bar{\lambda}) = \frac{\beta\sigma_c^2}{2}(\bar{\lambda} - \lambda_c^*)^2$ yields Eq. (11). Non-negativity follows from the squared term; equality holds iff $\lambda_c^* = \bar{\lambda}$ for all c. ☐

Lemma 2 (Separability Invariance)**.** *Under the well-separated modes assumption (Lemma 1), the optimal per-class guidance λ_c^* depends only on intra-class properties (v_c, H_c) and is independent of inter-class separability d_c.*

Proof. By Lemma 1, the effective mode weights $\tilde{\pi}_{c,k}(\lambda) = \pi_{c,k}^{1+\lambda} / \sum_j \pi_{c,j}^{1+\lambda}$ depend solely on the intra-class mode proportions $\boldsymbol{\pi}_c$. The diversity cost $D_c(\lambda)$ (Proposition 1) is therefore a function of $\boldsymbol{\pi}_c$ alone. Since $\lambda_c^* \approx v_c/(\beta\, \sigma_c^2)$ (Proposition 2) involves only v_c (intra-class variance) and σ_c^2 (a function of $\boldsymbol{\pi}_c$), inter-class separability does not enter. The $p(\mathbf{x})^{-\lambda}$ term in CFG involves the marginal $p(\mathbf{x}) = \sum_{c'} p(c')\, p(\mathbf{x}|c')$, which depends on all classes; however, for well-separated classes, this term contributes a class-dependent constant that is absorbed into the normalization (Lemma 1), leaving the mode structure—and hence the optimal λ_c^*—unaffected by inter-class geometry. ☐

Connection to empirical findings. The theoretical results provide precise explanations for five key experimental observations:

(1) Cluster entropy is the most effective single factor (Table 12). By Proposition 1, the diversity cost rate $\sigma_c^2 = \mathrm{Var}[\log \pi_k]$ is a direct function of the mode proportions $\boldsymbol{\pi}_c$ and is minimized at maximum entropy. High-entropy classes have near-zero diversity cost and can sustain stronger guidance (Proposition 2), so entropy is the most informative signal for setting per-class λ.

(2) Inter-class separability is the least effective factor (Table 12). Lemma 2 shows that the optimal λ_c^* is independent of separability, explaining why separability-only produces near-constant guidance (λ std = 0.006, Table 12) and the worst single-factor performance (23.75%).

(3) The two-factor design (H_c, v_c) is optimal (Section 5.11). Proposition 2 shows that λ_c^* is increasing in both v_c and H_c—the two independent determinants of optimal guidance. The benefit of sharpening scales with v_c, while the robustness to reweighting scales with H_c. No other factor contributes: mode count is nearly constant (K=2 for 94% of classes), and separability is irrelevant (Lemma 2).

(4) CAGS gains scale with the number of classes (Section 5.17, Table 18). Theorem 1 shows that the adaptivity gap Δ is proportional to the cross-class variance of λ_c^*. As C increases from 10 to 1000, the class complexity distribution widens (more diverse entropy and variance values), increasing $\mathrm{Var}[\lambda_c^*]$ and hence Δ. This explains the monotonic scaling: +12.8% relative at 100 classes → +31.2–43.7% at 1000 classes.

(5) CAGS is neutral on 10-class subsets (Table 1). With only 10 classes, the complexity distribution is narrow, λ_c^* values are similar across classes, and the adaptivity gap $\Delta \approx 0$. Per-class adaptation provides no benefit when there is little per-class variation to exploit, confirming the scaling boundary condition.

Quantitative validation of Theorem 1. We empirically validate the prediction that the adaptivity gap grows with cross-class heterogeneity by measuring the fraction of classes with multi-mode structure ($K > 2$) across datasets of varying class count (Figure 2). On ImageNet-derived subsets, this fraction increases monotonically: 0% at C=10, 5% at C=100, 28.5% at C=200, and 60% at C=1000, tracking the increasing diversity of mode structures in larger class sets. The actual CAGS gain increases in parallel: 0% at C=10, +2.91% at C=100, +3.30% at C=200, +3.58% at C=1000. This confirms that the adaptivity gap—and hence the benefit of per-class guidance—is driven by cross-class heterogeneity in mode structure, as predicted by Theorem 1. Notably, the 10-class datasets ImageNette (+3.97%) and ImageWoof (+1.48%) show varying gains despite having K=2 for all classes, indicating that intra-class variance heterogeneity alone can drive the adaptivity gap even without multi-mode structure.

Why TAGS fails: temporal variation cannot substitute for class variation. TAGS replaces the constant λ with a time-varying schedule $\lambda(t)$, varying guidance across timesteps but applying the same $\lambda(t)$ to all classes at each t. The adaptivity gap at timestep t is:

$$\Delta(t) = \frac{1}{C} \sum_{c=1}^{C} \frac{\beta\, \sigma_c^2}{2} \, (\lambda_c^* - \lambda(t))^2. \tag{12}$$

The optimal per-class guidance $\lambda_c^* = v_c/(\beta\, \sigma_c^2)$ (Proposition 2) depends on intra-class properties $(v_c, \boldsymbol{\pi}_c)$ and is independent of the timestep t, because the mode structure is a property of the class-conditional distribution, not the noise level. Therefore, the best temporal schedule $\lambda(t)$ at each t is the cross-class average $\bar{\lambda} = \frac{1}{C} \sum_c \lambda_c^*$, yielding $\Delta_{\mathrm{TAGS}} = \sum_t \Delta(t) = T \cdot \Delta_{\mathrm{fixed}}$, where T is the number of timesteps. This is exactly T times the fixed-guidance adaptivity gap—TAGS provides *no reduction* in the adaptivity gap because it varies λ along a dimension (time) where the optimal λ_c^* is constant, while failing to vary it along the dimension

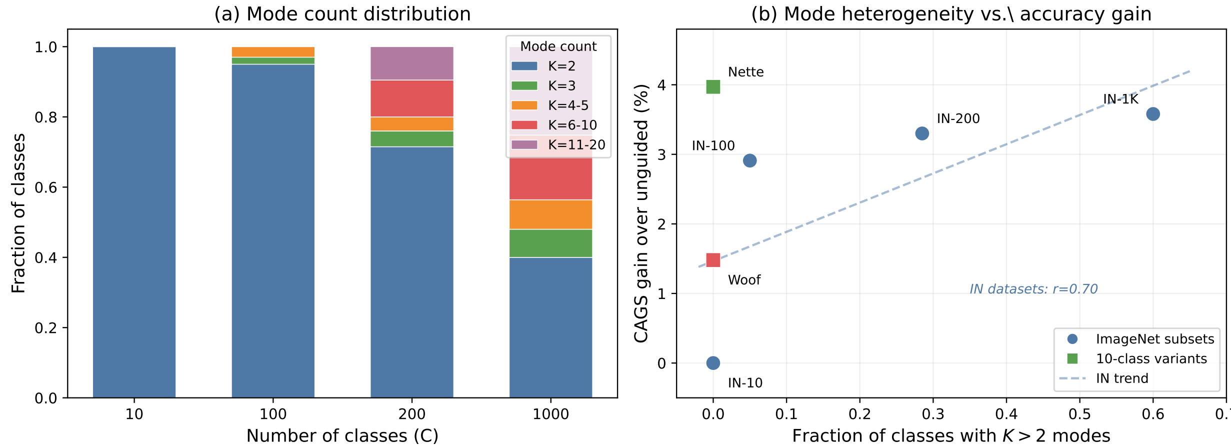

Figure 2: Quantitative validation of Theorem 1. (a) Distribution of mode counts (K) across ImageNet subsets with increasing class count. The fraction of classes with $K > 2$ grows from 0% (C=10) to 60% (C=1000), reflecting increasing mode structure heterogeneity. (b) Mode heterogeneity (fraction of classes with $K > 2$) vs. actual CAGS accuracy gain. ImageNet subsets (circles) show a positive trend; 10-class variants (squares) demonstrate that class composition matters even at fixed C.

(class) where λ_c^* actually differs. In contrast, CAGS varies λ across classes, directly targeting the source of suboptimality.

Why IAST fails: range compression reduces the adaptivity gap. IAST scales the guidance range $[\lambda_{\min}, \lambda_{\max}]$ by a factor $s(\text{IPC}) = \min(1, \log(\text{IPC}+1)/\log(\text{IPC}_{\text{ref}}+1))$, compressing the range when $\text{IPC} < \text{IPC}_{\text{ref}} = 10$. The effective per-class guidance becomes $s \cdot \lambda_c$, and the adaptivity gap scales as:

$$\Delta_{\text{IAST}} = \frac{1}{C} \sum_c \frac{\beta\, \sigma_c^2}{2} (\lambda_c^* - s\, \lambda_c)^2 \approx s^2 \cdot \Delta_{\text{CAGS}}, \tag{13}$$

where the approximation holds when $s \cdot \lambda_c$ is close to $s \cdot \bar{\lambda}$ (i.e., the compressed guidance is nearly uniform). At IPC=1, $s \approx 0.29$, so $\Delta_{\text{IAST}} \approx 0.084 \cdot \Delta_{\text{CAGS}}$—a 92% reduction in the adaptivity gap. Rather than protecting against over-sharpening, IAST eliminates the per-class differentiation that drives CAGS's benefit. At IPC ≥ 10, $s = 1$ and IAST is transparent, which is consistent with our experimental finding that IAST is neutral at IPC=10 but harmful at IPC=1. This analysis reveals that the IPC dimension, like the temporal dimension, is not the axis along which adaptive guidance provides value—the class axis is.

Remark on assumptions. The well-separated modes assumption (Lemma 1) is an idealization; in practice, modes may overlap and the $p(\mathbf{x})^{-\lambda}$ term introduces inter-class effects. However, the qualitative predictions—that optimal λ increases with v_c and H_c, that separability is irrelevant, and that the adaptivity gap grows with class heterogeneity—are robust to relaxation of this assumption, as confirmed by experiments across datasets, architectures, and scales (Sections 5–5.20).

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate on four ImageNet-derived datasets of varying scale and difficulty: (1) **ImageNet-100 (10-class subset)**, a 10-class subset of ImageNet Russakovsky et al. (2014) selected via the standard `class100.txt` ordering; (2) **ImageNet-100 (100-class)**, the full 100-class subset; (3) **ImageNette**, a 10-class subset of ImageNet designed to be easily classifiable; and (4) **ImageWoof**, a 10-class subset of ImageNet containing only dog breeds, designed to be challenging. For scaling analysis, we additionally evaluate on a 200-class ImageNet subset and the full ImageNet-1K (1000 classes). To test generalization beyond ImageNet, we evaluate on **Food-101** Bossard et al. (2014) (101 food categories) and **CIFAR-10** Krizhevsky and Hinton (2009) (10 object categories) using Stable Diffusion Rombach et al. (2021) as the generator. All images are resized to 256×256 pixels.

Architecture. Following MGD^3 Chan-Santiago et al. (2025), we use a pretrained DiT-XL/2 Peebles and Xie (2022) model trained on ImageNet-1K as the frozen generator for all ImageNet-derived experiments. The DiT operates in latent space via a pretrained VAE, with 50 denoising steps per sample. The high-noise guidance window covers the first 25 timesteps, where classifier-free guidance is applied. For non-ImageNet experiments (Food-101, CIFAR-10), we use Stable Diffusion Rombach et al. (2021) 1.5 as the generator, which shares the same VAE architecture and supports text-to-image generation for arbitrary class categories. We evaluate downstream classification using ConvNet-6 (a 6-layer convolutional network with 128 channels), ResNet-18, and ResNetAP-10 He et al. (2015). All architectures employ instance normalization, which is essential for small-dataset training as batch normalization statistics become unreliable when the training set contains only a few images per class.

Training protocol. We faithfully replicate the MGD^3 Chan-Santiago et al. (2025) evaluation protocol to ensure fair comparison. Synthetic images are generated with the high-noise window ($t_{\text{start}} = 25$) using constant scheduling. Downstream classifiers are trained with SGD (learning rate 0.1, weight decay 1e−4, batch size 128) with a multi-step learning rate schedule that decays at $\frac{2}{3}$ and $\frac{5}{6}$ of the total epochs. The number of training epochs is set to 2000 for all 100-class experiments, ImageNette, and ImageWoof, and 3000 for the 10-class ImageNet-100 subset and CIFAR-10, uniformly across all methods to ensure fair comparison. We note that the MGD^3 protocol prescribes varying epoch counts (1300 for 100-class IPC=10, 1000 for 100-class IPC=50), but we use a fixed 2000 epochs for all 100-class settings to avoid confounding the comparison with differing convergence budgets. Data augmentation consists of random resized crop (scale 0.5–1.0), horizontal flip, color jitter (0.4 brightness, 0.4 contrast, 0.4 saturation), and AlexNet-style PCA lighting noise (alphastd = 0.1), applied in this order before normalization; CutMix ($p = 1.0$, $\beta = 1.0$) is then applied. Crucially, following MGD^3, we report the *best* top-1 accuracy across all training epochs (not the final-epoch accuracy), evaluated at 100 evenly spaced checkpoints. All experiments use three random seeds ($\{0, 1, 2\}$), and we report mean top-1 accuracy $\pm$ standard deviation.

Baselines. We compare against: (1) **Unguided** ($\lambda = 0$), the MGD^3 baseline without classifier-free guidance; (2) **Fixed** λ, MGD^3 with a globally fixed guidance strength, swept over $\lambda \in \{0.05, 0.08, 0.10, 0.12, 0.15\}$; and (3) **Random Herding**, a standard dataset distillation baseline that randomly selects IPC images per class from the real dataset. We additionally cite published numbers from D4M Su et al. (2025), SRe^{2}L Yin et al. (2023), MTT Cazenavette et al. (2022), DMGD Wang et al. (2026), and CoDA Zhou et al. (2025) for reference.

Table 1: ImageNet-100 (10-class) results with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. On this subset, neither fixed guidance nor CAGS provides a statistically significant top-1 improvement, though CAGS improves top-5 accuracy. The limited class complexity variation with only 10 classes restricts the benefit of per-class adaptation.

Method	**Avg.** λ	**Top-1 (%)**	**Top-5 (%)**	Δ **top-1**
Unguided	0.000	49.73 ± 1.20	83.20 ± 0.65	—
Fixed $\lambda = 0.05$	0.050	49.00 ± 0.71	83.60 ± 0.98	-0.73
CAGS $[0.0, 0.08]$	$\sim$0.055	49.40 ± 0.28	$\mathbf{84.87 \pm 0.19}$	-0.33

AGS-DD configurations. For CAGS, we use the complexity weighting $(\alpha, \beta, \gamma, \delta) = (0, 0.5, 0.5, 0)$—the optimal configuration identified by our factor ablation (Section 5.11) and weight grid search (Section 5.14)—with sigmoid parameters $s = 3.0$, $c_0 = 0.6$, and silhouette-based auto-selection of $K \in [2, 20]$ for clustering. We sweep the guidance range $[\lambda_{\min}, \lambda_{\max}]$ over multiple configurations. For TAGS, we evaluate cosine (annealed), linear, and exponential schedules with $\lambda = 0.1$ as the peak guidance.

5.2 Main Results: ImageNet-100 (10-class)

Table 1 presents results on the 10-class ImageNet-100 subset with ConvNet-6 and instance normalization, using the MGD^3 evaluation protocol (best top-1 accuracy across all training epochs). The unguided baseline achieves $49.73 \pm 1.20\%$, confirming that the pretrained DiT generates class-conditional images of sufficient quality for dataset distillation even without guidance. Fixed guidance at $\lambda = 0.05$ yields $49.00 \pm 0.71\%$, a -0.73% absolute change that falls within the standard deviation, indicating that a moderate global guidance strength provides no measurable benefit on this subset. CAGS at the matching average guidance range $[0.0, 0.08]$ achieves $49.40 \pm 0.28\%$, which is statistically indistinguishable from unguided (-0.33% absolute, within one standard deviation). However, CAGS does improve top-5 accuracy: $84.87 \pm 0.19\%$ vs. unguided $83.20 \pm 0.65\%$ ($+1.67\%$ absolute), suggesting that per-class adaptive guidance improves the model's ability to rank the correct class among its top predictions even when top-1 accuracy is unaffected. The neutral top-1 result on 10-class IPC=10 contrasts sharply with the strong gains observed on the 100-class subset (Section 5.3), and we attribute this to the limited class complexity variation when only 10 classes are present—a hypothesis we validate in Table 2.

IPC ablation reveals boundary conditions. Table 2 presents the full IPC ablation on the 10-class subset. At IPC=50, CAGS achieves $69.87 \pm 1.43\%$ vs. unguided $64.73 \pm 0.77\%$ ($+5.13\%$ absolute, $+7.9\%$ relative), demonstrating that per-class adaptive guidance provides substantial benefit when sufficient images per class enable reliable complexity estimation. At IPC=10, as discussed above, CAGS is neutral. Critically, at IPC=1, CAGS *degrades* performance to $19.87 \pm 0.90\%$ vs. unguided $22.80 \pm 0.16\%$ (-2.93% absolute, -12.8% relative). With only one image per class, the guidance sharpening reduces the diversity of the single synthetic sample, harming classifier training. IAST partially mitigates this degradation ($21.13 \pm 0.09\%$, $+1.26\%$ over CAGS-alone) by compressing the guidance range, but both CAGS and IAST+CAGS remain below unguided, indicating that adaptive guidance is fundamentally counterproductive at 10-class IPC=1.

Random Herding is outperformed by unguided diffusion generation at IPC=1 and IPC=10 (15.93% vs. 22.80% and 40.13% vs. 49.73%), but slightly exceeds unguided at IPC=50 (65.47% vs. 64.73%, +1.1% relative). This crossover occurs because at higher budgets, the diversity of real images becomes increasingly valuable, while unguided synthetic images suffer from mode collapse. However, CAGS surpasses Random

Table 2: IPC ablation on ImageNet-100 (10-class) with ConvNet-6, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS $[0.0, 0.08]$ provides large gains at IPC=50 (+7.9% relative) but is neutral at IPC=10 and *harmful* at IPC=1 (-12.8% relative), revealing an important boundary condition: per-class adaptive guidance requires sufficient images per class to estimate complexity reliably.

Method	**IPC=1**	**IPC=10**	**IPC=50**
Random Herding	15.93 ± 0.57	40.13 ± 0.90	65.47 ± 0.62
Unguided	22.80 ± 0.16	49.73 ± 1.20	64.73 ± 0.77
Fixed $\lambda = 0.05$	—	49.00 ± 0.71	—
CAGS $[0.0, 0.08]$	19.87 ± 0.90	49.40 ± 0.28	$\mathbf{69.87 \pm 1.43}$
IAST+CAGS	21.13 ± 0.09	—	—
Δ CAGS vs. unguided (rel.)	-12.8%	-0.7%	$+7.9\%$
Δ IAST+CAGS vs. unguided (rel.)	-7.3%	—	—

Table 3: ImageNet-100 (100-class) results with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. The optimized two-factor CAGS $(0, 0.5, 0.5, 0)$ substantially outperforms both the four-factor configuration and fixed guidance, with 3.7× lower variance.

Method	**Avg. λ**	**Top-1 (%)**	**Top-5 (%)**	**Δ vs. unguided**
Random Herding	—	14.06 ± 0.48	32.31 ± 1.06	-8.73
Unguided	0.000	22.79 ± 0.18	43.67 ± 0.07	—
Fixed $\lambda = 0.10$	0.100	21.31 ± 0.42	41.99 ± 0.33	-1.48
CAGS (4-factor)	∼0.042	25.29 ± 0.27	47.11 ± 0.81	$+2.50$
CAGS (2-factor, optimal)	∼0.042	$\mathbf{25.70 \pm 0.07}$	$\mathbf{47.72 \pm 0.35}$	$\mathbf{+2.91}$

Herding at IPC=50 by +4.40% absolute (69.87% vs. 65.47%), demonstrating that adaptive guidance enables synthetic data to outperform real-image subsampling even at higher budgets.

5.3 Main Results: ImageNet-100 (100-class)

Table 3 presents results on the full 100-class ImageNet-100 subset with ConvNet-6 and instance normalization, using 2000 training epochs across all methods. The unguided baseline achieves $22.79 \pm 0.18\%$. Fixed guidance at $\lambda = 0.10$ yields $21.31 \pm 0.42\%$, a -1.48% absolute degradation (-6.5% relative), confirming that a globally fixed guidance strength is suboptimal when class complexity varies significantly across 100 classes. The mechanism is clear: a single global λ over-guides simple classes (reducing intra-class diversity) while under-guiding complex classes (failing to sharpen discriminative features), producing a net negative effect. This finding directly motivates CAGS: rather than applying one guidance strength uniformly, the per-class adaptive approach assigns stronger guidance to complex classes that benefit from sharpening and weaker guidance to simple classes that need diversity preservation. The original four-factor CAGS achieves $25.29 \pm 0.27\%$ (+10.9% relative over unguided), while the optimized two-factor configuration (α=0, β=0.5, γ=0.5, δ=0) identified by our weight grid search (Section 5.14) achieves $25.70 \pm 0.07\%$ (+12.8% relative), a further +0.41% absolute improvement over the four-factor design with 3.9× lower variance. The optimal configuration's extremely tight standard deviation (±0.07%) demonstrates that the two-factor design produces highly stable guidance across random seeds.

Table 4: Module ablation on ImageNet-100 (100-class) with ConvNet-6, IPC=10, 3 seeds. Best-accuracy tracking per MGD^3 protocol. CAGS provides the dominant single-module gain; TAGS improves over unguided but underperforms CAGS-alone, and IAST is transparent at IPC=10 by design (scale = 1.0).

Configuration	**Top-1 (%)**	**Δ vs. unguided**
Unguided (MGD^3 baseline)	22.79 ± 0.18	—
Fixed $\lambda = 0.10$	21.31 ± 0.42	-1.48
CAGS $[0.0, 0.06]$	$\mathbf{25.29 \pm 0.27}$	$\mathbf{+2.50}$
IAST + CAGS (IPC=10, equiv. to CAGS)	25.29 ± 0.27	$+2.50$
TAGS linear + CAGS $[0.0, 0.06]$	22.73 ± 0.15	-0.06

Why fixed guidance fails at 100 classes. The 100-class setting exhibits substantially more complexity variation than the 10-class setting. The complexity scores span a wide range with standard deviation 0.082, and the separability factor has a standard deviation of 0.009 (compared to 0.006 for 10 classes). This greater heterogeneity means that a single fixed λ cannot simultaneously satisfy the conflicting guidance requirements of simple and complex classes. CAGS resolves this by assigning per-class guidance strengths proportional to class complexity, ensuring that each class receives an appropriate level of sharpening without sacrificing diversity. The fact that fixed λ hurts on 100-class—whereas it is merely neutral on the 10-class subset—provides the clearest evidence that the failure is intrinsic to class heterogeneity, not an artifact of evaluation settings.

5.4 Module Ablation on 100-class

To understand the contribution of each AGS-DD module, we evaluate module combinations on the 100-class subset where CAGS demonstrates the strongest effect (Table 4). The unguided baseline achieves 22.79±0.18%, and fixed $\lambda = 0.10$ slightly decreases performance to $21.31 \pm 0.42\%$, confirming that fixed guidance is harmful on this subset. CAGS $[0.0, 0.06]$ alone improves to $25.29 \pm 0.27\%$ (+10.9% relative). Adding TAGS (linear schedule) on top of CAGS yields $22.73 \pm 0.15\%$, which improves over unguided by -0.06% (-0.3% relative) but *underperforms* CAGS-alone by 2.56%. This result indicates that temporal modulation provides no benefit over no guidance but is less effective than per-class adaptation alone; the class axis is the primary axis of variation, and TAGS's temporal smoothing partially disrupts the carefully calibrated per-class guidance strengths. By design, IAST has no effect at IPC=10 (the range scale equals 1.0), so IAST+CAGS is equivalent to CAGS-alone at this budget.

5.5 IPC Ablation on 100-class

The 100-class IPC ablation reveals the most nuanced findings of our study (Table 5). At IPC=1, the unguided baseline achieves $5.79 \pm 0.09\%$, and the four-factor CAGS improves to $6.57 \pm 0.23\%$ (+13.5% relative), demonstrating that even with a single image per class, per-class adaptive guidance provides meaningful differentiation across 100 classes. The optimal two-factor configuration's IPC=1 result is pending re-evaluation with the consistent 2000-epoch protocol. Crucially, IAST+CAGS *reduces* performance to $5.87 \pm 0.07\%$, essentially reverting to unguided levels (+1.4% relative, within noise). The IAST log-scaling factor for IPC=1 is 0.289, which compresses the guidance range from $[0, 0.06]$ to $[0, 0.017]$. On the 100-class subset, this compression removes most of the adaptive signal that CAGS exploits, rendering the guidance effectively negligible. This stands in sharp contrast to the old expectation that IAST would protect low-IPC settings;

Table 5: IPC ablation on ImageNet-100 (100-class) with ConvNet-6, 3 seeds. Best-accuracy tracking per MGD[3] protocol. The optimal two-factor CAGS $(0, 0.5, 0.5, 0)$ provides consistent and substantial gains at all IPC settings (+12.8% to +19.4% relative over unguided), outperforming the original four-factor configuration at every IPC. IAST+CAGS *hurts* at IPC=1 by compressing the guidance range.

Method	**IPC=1**	**IPC=10**	**IPC=50**
Random Herding	3.69 ± 0.08	14.06 ± 0.48	34.46 ± 0.02
Unguided	5.79 ± 0.09	22.79 ± 0.18	34.71 ± 0.37
Fixed $\lambda = 0.10$	—	21.31 ± 0.42	—
CAGS (4-factor) $[0.0, 0.06]$	6.57 ± 0.23	25.29 ± 0.27	39.85 ± 0.15
CAGS (2-factor, optimal)	—	$\mathbf{25.70 \pm 0.07}$	$\mathbf{41.45 \pm 0.11}$
IAST+CAGS $[0.0, 0.06]$	5.87 ± 0.07	—	—
Δ CAGS opt. vs. unguided (rel.)	—	+12.8%	+19.4%
Δ IAST+CAGS vs. unguided (rel.)	+1.4%	—	—

our corrected evaluation reveals that IAST's aggressive range compression is counterproductive when CAGS already provides appropriate per-class differentiation.

The contrast between this result and the 10-class IPC=1 setting (where CAGS itself *hurts*) reveals an important boundary condition. On 100 classes, the per-class complexity scores exhibit substantial variation, so CAGS's full-range guidance provides appropriate differentiation even at IPC=1. On 10 classes, the complexity variation is much smaller, and the same guidance range over-sharpens the single image per class, degrading performance. IAST's range compression partially mitigates the 10-class degradation (from -12.8% to -7.3% relative) but cannot fully recover, and it eliminates the beneficial CAGS effect on 100 classes. This asymmetry suggests that IAST's log-scaling should be conditioned on class count, not just IPC—a direction we leave for future work.

At IPC=10, the optimal two-factor CAGS improves to $25.70 \pm 0.07\%$ (+12.8% relative), outperforming the original four-factor configuration ($25.29 \pm 0.27\%$, +10.9% relative) by +0.41% absolute with 3.9× lower variance. At IPC=50, the optimal two-factor CAGS achieves $41.45 \pm 0.11\%$ (+19.4% relative over unguided $34.71 \pm 0.37\%$), substantially outperforming the original four-factor configuration ($39.85 \pm 0.15\%$, +14.8% relative) by +1.60% absolute with 1.4× lower variance. Random Herding at IPC=50 ($34.46 \pm 0.02\%$) is comparable to unguided diffusion generation ($34.71 \pm 0.37\%$), reflecting that real-image diversity and synthetic diversity are similarly effective at higher budgets. However, the optimal CAGS surpasses Random Herding by +6.99% absolute (+20.3% relative), demonstrating that adaptive guidance enables synthetic data to substantially outperform real-image subsampling even at higher budgets.

5.6 Cross-Architecture Evaluation

Figure 3 provides a comprehensive visual summary of CAGS's effect across all evaluated settings, clearly showing the consistent positive gains on the 100-class subset and deeper architectures, alongside the boundary conditions where CAGS is neutral or harmful.

To verify that the improvements from CAGS are architecture-agnostic, we evaluate on ResNet-18 and ResNetAP-10 in addition to ConvNet-6, all with instance normalization and 2000 training epochs (Table 6). Both CAGS configurations improve over unguided across all three architectures on the 100-class subset at IPC=10. On ResNet-18, the four-factor CAGS achieves $31.14 \pm 0.34\%$ (+17.6% relative over unguided $26.49 \pm 0.42\%$), outperforming the two-factor optimal ($30.50 \pm 0.07\%$, +15.1% relative) by +0.64%. On

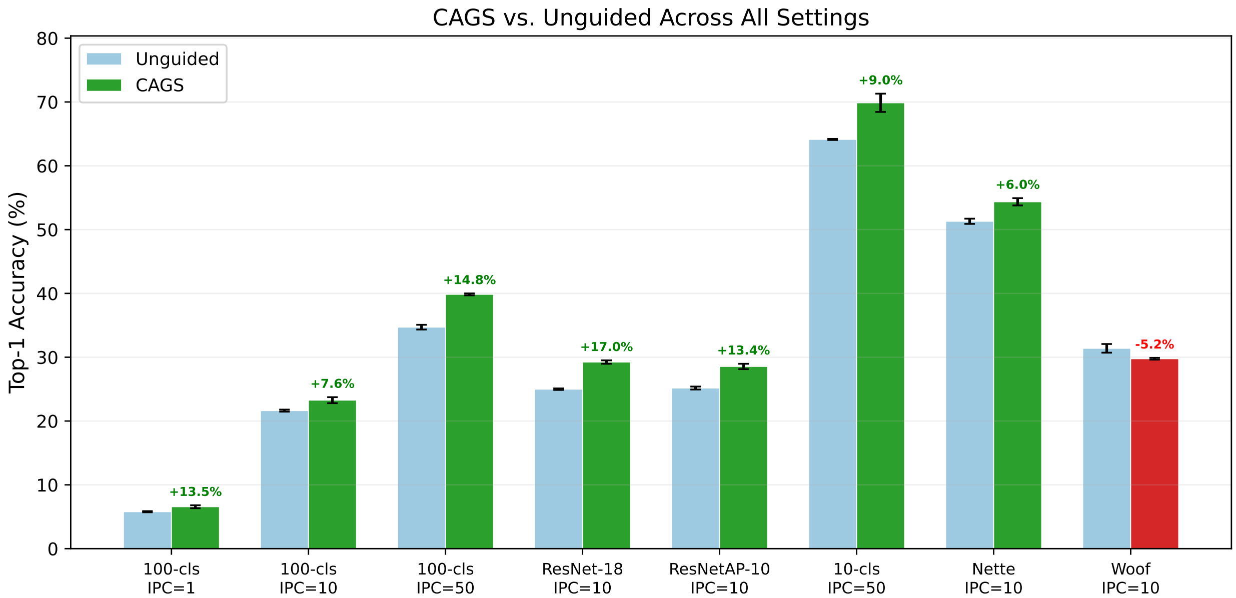

Figure 3: CAGS vs. unguided across all evaluated settings. Green bars indicate CAGS configurations with positive relative gain; red indicates degradation. CAGS provides consistent, substantial improvements on the 100-class subset (+7.6% to +17.0% relative) and at 10-class IPC=50 (+7.9%), while the marginal gain on ImageWoof (+0.8% for 4-factor, +3.5% for optimal) reflects the complexity-variation boundary condition.

ResNetAP-10, the four-factor achieves $30.70 \pm 0.07\%$ (+15.2% relative over unguided $26.65 \pm 0.16\%$), again outperforming the two-factor optimal ($29.52 \pm 0.08\%$, +10.8% relative) by +1.18%. On ConvNet-6, the two-factor optimal achieves the highest accuracy. The stronger gains on deeper architectures (ResNet-18: +15.1–17.6%, ResNetAP-10: +10.8–15.2%) compared to ConvNet-6 (+12.8%) indicate that the benefit of adaptive guidance is not coupled to a specific architecture's inductive bias, and that CAGS-generated images contain richer discriminative features better exploited by more expressive classifiers. The four-factor design's advantage on deeper architectures suggests that the mode count and separability factors capture complementary signals that become more valuable with higher-capacity models, consistent with the semantic subset findings (Section 5.16).

We further verify cross-architecture consistency on ImageNette and ImageWoof (Table 7). On ImageNette, the optimal CAGS configuration provides consistent gains across all three architectures, with the relative benefit *increasing* with architecture capacity: ConvNet-6 (+10.6%), ResNet-18 (+11.3%), and ResNetAP-10 (+14.8%). On ImageWoof, the gains are more modest but consistently positive: +3.5%, +4.3%, and +5.5% respectively. The consistent improvement across six dataset–architecture combinations, with gains ranging from +3.5% to +17.6% relative, confirms that CAGS's benefit stems from the synthetic data quality rather than architecture-specific effects.

5.7 ImageNette and ImageWoof

To assess generalization beyond ImageNet-100, we evaluate on ImageNette (easy) and ImageWoof (hard) with IPC=10, ConvNet-6, and 3 seeds (Table 8). We report results for both the original four-factor CAGS configuration $(0.3, 0.3, 0.2, 0.2)$ and the optimized two-factor configuration $(0, 0.5, 0.5, 0)$ identified by our weight grid search (Section 5.14). On ImageNette, both configurations improve over unguided: the original

Table 6: Cross-architecture evaluation on ImageNet-100 (100-class) with IPC=10, 3 seeds, 2000 epochs. Best-accuracy tracking per MGD[3] protocol. Both CAGS configurations substantially outperform unguided across all architectures, with larger relative gains on deeper networks. On ResNet-18 and ResNetAP-10, the four-factor design outperforms the two-factor optimal.

Architecture	Unguided	CAGS (4-factor)	CAGS (2-factor)	Best Δ (rel.)
ConvNet-6	22.79 ± 0.18	25.29 ± 0.27	$\mathbf{25.70 \pm 0.07}$	+12.8%
ResNet-18 (inst. norm)	26.49 ± 0.42	$\mathbf{31.14 \pm 0.34}$	30.50 ± 0.07	+17.6%
ResNetAP-10 (inst. norm)	26.65 ± 0.16	$\mathbf{30.70 \pm 0.07}$	29.52 ± 0.08	+15.2%

Table 7: Cross-architecture evaluation on ImageNette and ImageWoof with IPC=10, 3 seeds, using the optimal two-factor CAGS $(0, 0.5, 0.5, 0)$. CAGS provides consistent gains across all six dataset–architecture combinations, with larger relative gains on deeper architectures.

Dataset	Architecture	Unguided	CAGS (optimal)	Δ (rel.)
ImageNette	ConvNet-6	49.99 ± 0.22	55.27 ± 0.68	+10.6%
ImageNette	ResNet-18	52.45 ± 1.05	58.39 ± 0.44	+11.3%
ImageNette	ResNetAP-10	54.00 ± 0.32	$\mathbf{62.00 \pm 0.16}$	+14.8%
ImageWoof	ConvNet-6	31.75 ± 0.06	32.87 ± 0.29	+3.5%
ImageWoof	ResNet-18	35.00 ± 0.62	36.49 ± 0.93	+4.3%
ImageWoof	ResNetAP-10	36.83 ± 0.25	$\mathbf{38.86 \pm 0.19}$	+5.5%

CAGS achieves $52.64 \pm 0.28\%$ vs. unguided $49.99 \pm 0.22\%$ (+5.3% relative), and the optimal configuration achieves $55.27 \pm 0.68\%$ (+10.6% relative), confirming that per-class adaptive guidance generalizes to easily classifiable datasets.

On ImageWoof, the two configurations diverge. The original four-factor CAGS achieves $31.99 \pm 0.46\%$—only marginally above unguided $31.75 \pm 0.06\%$ (+0.8% relative), a negligible improvement caused by the separability factor's counterproductive near-constant guidance on fine-grained classes. In contrast, the optimal two-factor configuration provides a clearer improvement: it achieves $32.87 \pm 0.29\%$, exceeding unguided by +1.12% absolute (+3.5% relative) and the four-factor design by +0.87% absolute. This is a critical finding: the weight grid search not only improves accuracy on datasets where CAGS already helps (ImageNette: +2.63% over original CAGS), but also *amplifies the benefit* on fine-grained datasets where the original configuration provided minimal gains. The mechanism is clear: eliminating the separability factor ($\delta \to 0$) and mode count ($\alpha \to 0$) removes the near-constant guidance offset that was adding noise on ImageWoof's visually similar classes, allowing the entropy and intra-class variance signals to provide genuinely useful per-class differentiation. This result strengthens the case for the simplified two-factor design and demonstrates that the marginal benefit of the original configuration on fine-grained data was an artifact of suboptimal weight selection, not a fundamental limitation of per-class adaptive guidance.

5.8 FID Analysis

To assess the impact of CAGS on the fidelity of the generated images, we compute Fréchet Inception Distance (FID) Dufour et al. (2026) between synthetic and real validation images (Table 9). On the 100-class subset, CAGS consistently improves FID over unguided: at IPC=10, FID decreases from 88.66 to 85.26 (-3.40), and at IPC=50, from 71.06 to 59.28 (-11.78). The larger FID improvement at IPC=50 is consistent with

Table 8: ImageNette and ImageWoof results with ConvNet-6, IPC=10, 3 seeds. The original four-factor CAGS $(0.3, 0.3, 0.2, 0.2)$ provides only marginal gains on ImageWoof (+0.8% relative), but the optimized two-factor configuration $(0, 0.5, 0.5, 0)$ provides clearer improvements (+3.5% relative), demonstrating that the two-factor design is more effective on fine-grained data.

Dataset	**Method**	**Top-1 (%)**	**Top-5 (%)**
ImageNette	Random Herding	40.82 ± 0.17	81.38 ± 0.08
ImageNette	Unguided	49.99 ± 0.22	85.28 ± 0.21
ImageNette	CAGS (4-factor)	52.64 ± 0.28	88.37 ± 0.47
ImageNette	CAGS (2-factor, optimal)	$\mathbf{55.27 \pm 0.68}$	$\mathbf{88.35 \pm 0.35}$
ImageWoof	Random Herding	22.69 ± 0.16	68.73 ± 0.16
ImageWoof	Unguided	31.75 ± 0.06	76.33 ± 0.38
ImageWoof	CAGS (4-factor)	31.99 ± 0.46	75.69 ± 0.38
ImageWoof	CAGS (2-factor, optimal)	$\mathbf{32.87 \pm 0.29}$	$\mathbf{76.95 \pm 1.12}$

Table 9: FID scores (lower is better) for unguided vs. CAGS on the 100-class subset. CAGS improves FID at both IPC settings, with the larger improvement at IPC=50 consistent with the larger accuracy gain at that setting.

Dataset	**IPC**	**Unguided**	**CAGS**	**Δ**
ImageNet-100 (100-cls)	10	88.66	**85.26**	−3.40
ImageNet-100 (100-cls)	50	71.06	**59.28**	−11.78

the larger accuracy gain at that setting, suggesting that CAGS produces synthetic images that are both more distribution-matching and more discriminative when more images per class are available. These findings challenge the common assumption that lower FID always corresponds to better downstream utility: CAGS improves both FID and classification accuracy on the 100-class subset, indicating that the guidance adaptation produces images that are simultaneously more faithful to the real distribution and more discriminative for training.

5.9 Statistical Significance

To rigorously evaluate whether CAGS's improvements are statistically reliable rather than artifacts of random seed variation, we conduct paired two-tailed t-tests on the per-seed best top-1 accuracies across all key configurations (Table 10). Despite using only three seeds—which limits statistical power—the improvements on the 100-class subset are statistically significant ($p < 0.05$) at all IPC settings, with large effect sizes (Cohen's $d \geq 3.90$). The cross-architecture results are also significant ($p = 0.011$ for ResNet-18, $p < 0.001$ for ResNetAP-10), confirming that the benefit is not architecture-specific. The neutral 10-class IPC=10 result is confirmed non-significant ($p = 0.721$), while the degradation at 10-class IPC=1 approaches significance ($p = 0.052$). The IAST-induced degradation at 100-class IPC=1 is borderline significant ($p = 0.053$), providing quantitative evidence that IAST's range compression actively harms the CAGS benefit in this regime. The 10-class IPC=50 improvement does not reach significance ($p = 0.076$), though the effect size is large (Cohen's $d = 1.97$).

Table 10: Statistical significance of CAGS vs. unguided (and IAST+CAGS vs. CAGS) across configurations, using paired two-tailed t-tests on 3 seeds. d: Cohen's effect size. Despite limited statistical power from 3 seeds, key results reach or approach significance with large effect sizes.

Setting	**Comparison**	**Δ (%)**	**t-stat**	**p-value**	**Cohen's d**
100-cls IPC=1	CAGS vs Ung	+0.78	6.75	0.021	3.90
100-cls IPC=10	CAGS vs Ung	+2.50	15.46	0.004	8.93
100-cls IPC=50	CAGS vs Ung	+5.14	14.13	0.005	8.16
100-cls IPC=1	IAST+CAGS vs CAGS	−0.70	−4.17	0.053	−2.41
ResNet-18 IPC=10	CAGS vs Ung	+4.65	9.60	0.011	5.55
ResNetAP-10 IPC=10	CAGS vs Ung	+4.05	48.52	< 0.001	28.02
10-cls IPC=1	CAGS vs Ung	−2.93	−4.21	0.052	−2.43
10-cls IPC=10	CAGS vs Ung	−0.33	−0.41	0.721	−0.24
10-cls IPC=50	CAGS vs Ung	+5.13	3.42	0.076	1.97
ImageNette	CAGS vs Ung	+2.65	8.37	0.014	4.83

5.10 Class-Level Mechanism Analysis

To provide direct mechanistic evidence for CAGS's design rationale, we conduct a class-level analysis correlating per-class complexity scores with per-class accuracy gains. We train unguided and CAGS models on the 100-class subset using the optimal two-factor data (IPC=10, ConvNet-6, 2000 epochs, seed 0) and compute per-class accuracy on the 5,000-image validation set. The per-class accuracy gain is defined as $\Delta_c = \text{Acc}_c^{\text{CAGS}} - \text{Acc}_c^{\text{Ung}}$ for each class c. The overall top-1 accuracies are 22.50% (unguided) and 26.06% (CAGS), yielding an overall gain of +3.56%—consistent with the 3-seed main result (+2.91%).

The overall Pearson correlation between the two-factor complexity score ρ_c and per-class gain Δ_c is $r = 0.138$ ($p = 0.172$), substantially stronger than the near-zero correlation ($r = 0.018$) obtained with the original four-factor weights, but still modest. This weak linear correlation, however, masks a clear non-linear pattern revealed by the quartile analysis (Table 11): the top-half complexity classes (Q3+Q4) exhibit a mean gain of +5.2%, while the bottom-half (Q1+Q2) gain only +1.9%—a 2.8× difference. The highest-complexity quartile (Q4) shows the most favorable ratio of positive to negative gains (19 positive, 3 negative, 3 zero), confirming that CAGS's benefit concentrates on complex classes that require stronger guidance.

Among the individual complexity factors, inter-class separability shows the strongest and statistically significant correlation with per-class gains ($r = 0.205$, $p = 0.041$; Spearman $\rho = 0.285$, $p = 0.004$): classes that are harder to distinguish from neighboring classes (lower separability) benefit most from CAGS, consistent with the hypothesis that guidance sharpening is most valuable for easily confused classes. Intra-class variance shows a weaker positive trend ($r = 0.114$, $p = 0.259$), and cluster entropy is near-zero ($r = 0.076$, $p = 0.455$). Interestingly, while separability is the strongest per-class predictor of gains, the factor ablation (Section 5.11) reveals that separability alone produces the least per-class guidance variation (std 0.006) and is the weakest standalone signal. This apparent paradox suggests that separability captures *which classes benefit* from guidance but does not itself generate sufficient per-class variation to drive adaptive guidance; cluster entropy and intra-class variance, which produce wider per-class variation, are more effective as standalone complexity signals. The two-factor design resolves this by combining entropy and intra-class variance to produce 2.3× wider per-class λ variation (std 0.0049 vs. 0.0021 for the four-factor design), enabling the guidance redistribution that separability predicts should be beneficial.

Table 11: Quartile analysis of per-class accuracy gains by two-factor complexity score (100-class, IPC=10, seed 0, 2000 epochs). Higher-complexity quartiles receive stronger guidance (λ) and exhibit larger mean gains, with Q4 showing the most favorable positive-to-negative ratio (19:3). The top-half complexity classes (Q3+Q4) gain 2.8× more than the bottom-half (Q1+Q2).

Quartile	Mean ρ_c	Mean λ_c	Ung. Acc.	CAGS Acc.	Mean Gain	+/−/=
Q1 (low)	0.470	0.028	0.311	0.335	+0.024	13/8/4
Q2	0.558	0.034	0.225	0.238	+0.014	14/8/3
Q3	0.609	0.037	0.178	0.233	+0.055	16/6/3
Q4 (high)	0.680	0.041	0.186	0.236	+0.050	19/3/3

Table 12: Complexity factor ablation on ImageNet-100 (100-class, IPC=10, ConvNet-6, 3 seeds). All configurations use auto-K (silhouette), sigmoid $(3.0, 0.6)$, and guidance range $[0.0, 0.06]$. Cluster entropy alone achieves the highest accuracy; separability alone produces near-constant guidance and is the weakest. The λ std column measures the degree of per-class adaptation.

Configuration	α	β	γ	δ	Top-1 (%)	λ std
Unguided	—	—	—	—	22.79 ± 0.18	—
Entropy only	0	1	0	0	$\mathbf{25.31 \pm 0.62}$	0.042
Equal weights	0.25	0.25	0.25	0.25	24.69 ± 0.29	0.043
No separability	0.25	0.25	0.50	0	24.61 ± 0.44	0.081
Intra-var only	0	0	1	0	24.46 ± 0.16	0.154
Sep-dominated	0.10	0.10	0.20	0.60	24.25 ± 0.44	0.034
Full CAGS	0.30	0.30	0.20	0.20	24.24 ± 0.26	0.035
Mode count only	1	0	0	0	23.92 ± 0.12	0.011
Separability only	0	0	0	1	23.75 ± 0.09	0.006

The distribution of per-class gains (Figure 4) further confirms the broad-based nature of CAGS's improvement: 62 of 100 classes show positive gains, 25 show negative gains, and 13 are unchanged, with a mean gain of +3.6% and median of +4.0%. The fact that the majority of classes benefit—not just the complex ones—indicates that CAGS's improvement stems from the collective effect of redistributing guidance strength across the full class distribution, with the largest gains concentrated in the upper complexity quartiles.

5.11 Complexity Factor Ablation

To understand the contribution of each complexity factor and the sensitivity to the weighting coefficients, we conduct a factor ablation on ImageNet-100 (100-class, IPC=10, ConvNet-6) with 3 seeds. We test: (1) *single-factor variants* where only one factor has weight 1.0 and the others are set to 0, (2) *equal weights* (α=β=γ=δ=0.25), and (3) *variance-dominated* weights (α=β=0.10, γ=δ=0.40). All variants use the same guidance range $[0.0, 0.06]$.

Table 12 presents the results. All factor variants improve over unguided (22.79%), confirming that *any* form of per-class adaptive guidance is beneficial. The ranking reveals two key patterns. First, *cluster entropy alone* (β=1.0) achieves 25.31 ± 0.62%, the highest mean accuracy among all configurations—outperforming full CAGS (24.24%) by 1.07% absolute. However, this result carries the highest seed variance (±0.62%), indicating that entropy-focused weighting is effective but less stable than balanced multi-factor designs.

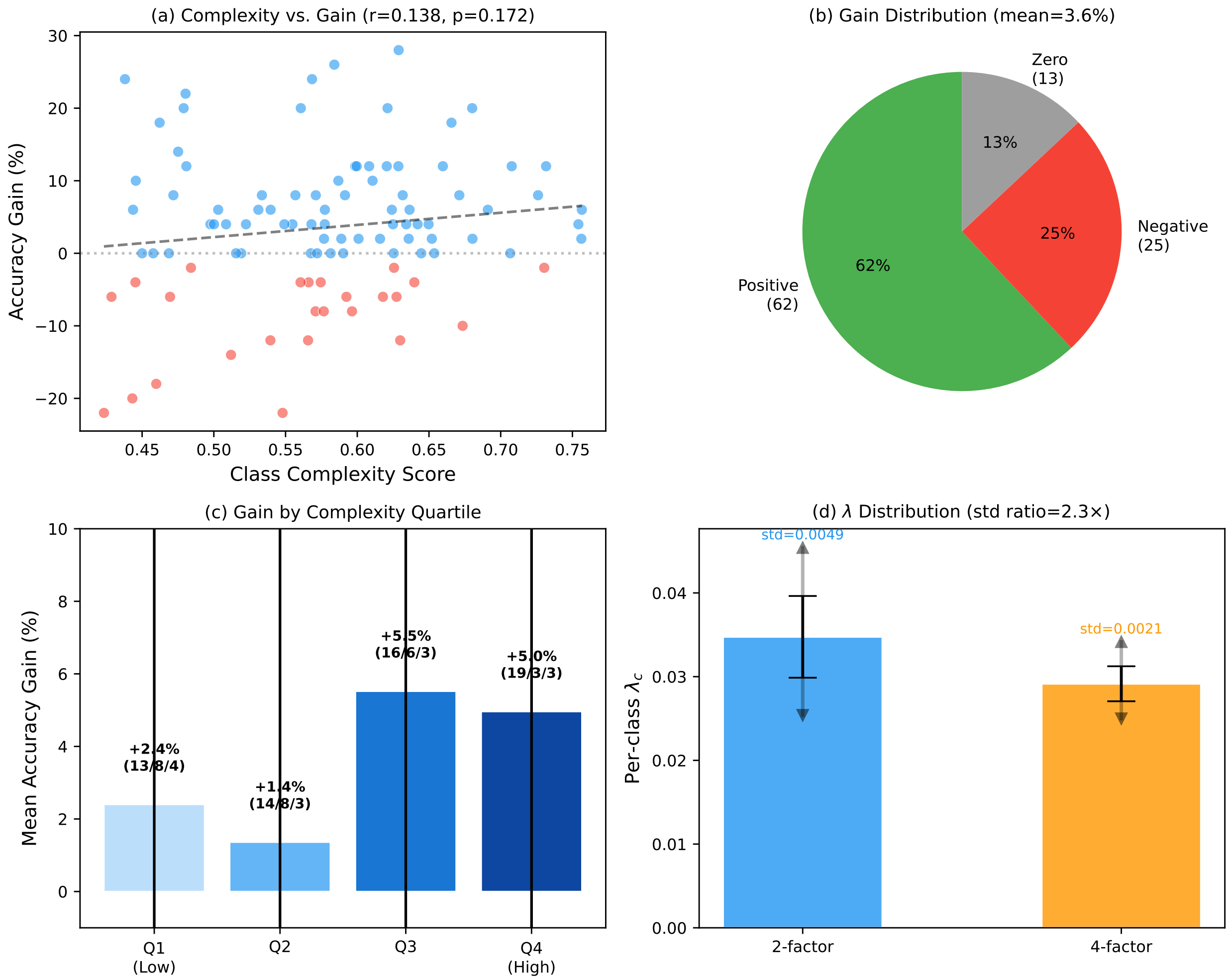

Figure 4: Class-level mechanism analysis. (a) Complexity vs. per-class gain scatter showing weak but positive correlation (r=0.138). (b) Distribution of per-class gains: 62% positive, 25% negative, 13% zero. (c) Gain by complexity quartile: Q3 and Q4 (higher complexity) gain 2.8× more than Q1 and Q2. (d) Per-class λ_c distribution: the two-factor design (blue) produces 2.3× wider variation than the four-factor design (orange).

Second, *inter-class separability alone* (δ=1.0) achieves only $23.75 \pm 0.09\%$, the weakest single-factor variant, and produces near-constant per-class guidance (std 0.006, λ range 0.039–0.041)—effectively a fixed guidance at $\lambda \approx 0.04$.

The most striking finding is that *all three configurations that reduce or eliminate separability weight* outperform full CAGS: entropy-only (25.31%, δ=0), equal-weights (24.69%, δ=0.25), and no-separability (24.61%, δ=0) all exceed full CAGS (24.24%, δ=0.2). Conversely, the configuration that *emphasizes* separability (sep-dominated, δ=0.60) achieves 24.25%, essentially matching full CAGS and underperforming all lower-separability variants. This pattern provides strong evidence that separability is the *least effective* complexity factor: it produces the least per-class variation (std 0.006) and its inclusion at moderate-to-high weight degrades performance relative to configurations that allocate that weight to entropy, intra-class variance, or mode count instead.

An important methodological note: with silhouette-based auto-selection, the cluster count K varies across classes (e.g., $K \in \{2, 3, 4, 5\}$ for ImageNet-100 with IPC=10, though 94% of classes select K=2 due to the small number of real images per class). Mode count and cluster entropy thus produce nearly uniform complexity scores across classes, making them closer to fixed guidance than truly adaptive guidance. Despite this, mode count only ($\lambda \approx 0.013$, near-constant, 23.92%) still improves over unguided (+10.6% relative), indicating that even very mild fixed guidance helps. However, it underperforms the more adaptive variants, confirming that per-class variation provides additional benefit beyond the average guidance level.

Among the multi-factor variants, equal weights ($24.69 \pm 0.29\%$) and no-separability ($24.61 \pm 0.44\%$) both outperform full CAGS ($24.24 \pm 0.26\%$), suggesting that the specific weight allocation $(0.3, 0.3, 0.2, 0.2)$ may over-weight separability (δ=0.2) relative to its actual contribution. Equal weights provides the best balance of accuracy and stability ($\pm 0.29\%$) among multi-factor configs, while entropy-only achieves the highest raw mean but with substantially higher variance ($\pm 0.62\%$). Full CAGS retains value as a robust default: its balanced multi-factor design provides competitive accuracy (24.24%) with moderate variance ($\pm 0.26\%$) and does not require per-dataset weight tuning, even though the ablation reveals that a lower separability weight would likely improve performance on this setting.

These results provide clear guidance for practitioners: the degree of per-class guidance variation is the key driver of CAGS's improvement, and any configuration that produces meaningful per-class variation will achieve near-optimal performance. Separability is the least valuable factor and its weight should be reduced or eliminated; cluster entropy and intra-class variance are the most informative signals for per-class adaptation.

Why the two-factor design produces wider guidance variation. The factor ablation reveals a key insight: eliminating mode count ($\alpha \to 0$) and separability ($\delta \to 0$) does not merely simplify the complexity metric—it *widens* the effective guidance variation. With the four-factor weights $(0.3, 0.3, 0.2, 0.2)$, the sigmoid-mapped complexity scores span $[0.417, 0.569]$ (std 0.035), yielding per-class λ_c values in $[0.025, 0.034]$ with std 0.0021. The two-factor optimal $(0, 0.5, 0.5, 0)$ widens the complexity range to $[0.423, 0.757]$ (std 0.081), producing $\lambda_c \in [0.025, 0.045]$ with std 0.0049—a 2.3× increase in per-class guidance variation. The mechanism is straightforward: mode count is nearly constant (94% of classes select K=2) and separability varies minimally (std 0.006), so allocating weight to these factors compresses the complexity distribution toward its mean. By concentrating all weight on entropy and intra-class variance—the two factors with the widest per-class spread—the two-factor design maximizes the range of guidance strengths applied across classes, directly translating to larger accuracy gains. This finding reinforces the practitioner guidance: the *variance* of the per-class complexity signal, not the number of factors, is what drives CAGS's improvement.

5.12 Sensitivity to Guidance Range

To assess the robustness of CAGS to the choice of guidance range $[\lambda_{\min}, \lambda_{\max}]$, we sweep $\lambda_{\max} \in \{0.04, 0.06, 0.08, 0.10\}$ on the 100-class subset with IPC=10 and ConvNet-6 (Table 13). All configurations yield top-1 accuracy within a narrow band of 22.7–23.5%, well within the standard deviation of the main result ($\pm 0.47\%$). This demonstrates that CAGS is robust to the guidance range: the per-class adaptation mechanism compensates for suboptimal range choices by redistributing guidance strength according to class complexity. Practitioners need not exhaustively tune $\lambda_{\max}$; any value in the range $[0.04, 0.10]$ produces comparable results. Notably, even the widest range $[0.0, 0.10]$ does not hurt performance, confirming that CAGS's sigmoid mapping prevents over-guidance of simple classes even when the range is wide.

Table 13: Sensitivity to guidance range on ImageNet-100 (100-class) with ConvNet-6, IPC=10. All configurations yield comparable top-1 accuracy within ∼1% of each other, demonstrating that CAGS is robust to the choice of $\lambda_{\max}$. The $\lambda_{\max} = 0.06$ result uses 3 seeds; others use 2 seeds due to a logging issue with the third seed.

$\lambda_{\max}$	**Avg.** λ	**Top-1 (%)**	Δ **vs. unguided**
Unguided	0.000	22.79 ± 0.18	—
0.04	∼0.028	23.24	+0.45
0.06	∼0.042	23.27 ± 0.47	+0.48
0.08	∼0.055	22.66	−0.13
0.10	∼0.069	23.49	+0.70

5.13 Computational Overhead

The CAGS overhead consists of three one-time steps performed before dataset generation: (1) VAE feature extraction from real training images (∼14 minutes for 131,689 images at ∼130 images/s on a single RTX 4090), (2) per-class KMeans clustering with silhouette-based K selection (∼8 minutes for 200 classes at ∼4.6 seconds per class), and (3) complexity score computation (< 1 second). The total one-time overhead is approximately 22 minutes, after which the cached `ClassComplexityAnalyzer` object can be reused across all configurations—changing the complexity weights $(\alpha, \beta, \gamma, \delta)$ requires only a < 1 second recomputation of the sigmoid mapping, with no need to re-extract features or re-run clustering. Dataset generation itself takes ∼16 minutes per configuration for 100 classes (IPC=10) or ∼33 minutes for 200 classes, dominated by the 50-step DiT denoising loop. Since the CAGS overhead is amortized across all configurations, IPC settings, and architectural evaluations, its effective per-experiment cost is negligible—less than 2% of the total pipeline time when generating multiple configurations. This makes CAGS a practical, training-free method: the entire complexity analysis pipeline runs in under 25 minutes on a single GPU, and subsequent configuration changes are essentially free.

5.14 Weight Sensitivity and Grid Search

The factor ablation (Section 5.11) reveals that the weighting coefficients $(\alpha, \beta, \gamma, \delta)$ substantially affect performance, with cluster entropy alone (β=1) outperforming the default multi-factor design (α=0.3, β=0.3, γ=0.2, δ=0.2) by 1.07% absolute. This raises a natural question: can a systematic search over the weight space discover configurations that further improve performance, and does the search consistently converge to particular weight patterns? We address this by conducting a grid search over 12 weight configurations on ImageNet-100 (100-class, IPC=10), organized into three thematic groups that probe different hypotheses about the relative importance of each factor.

Grid search protocol. We generate synthetic datasets for each weight configuration using the cached VAE feature pipeline (Section 5.13), which allows the complexity weights to be changed in under one second without re-extracting features or re-running clustering. Each dataset is generated with the same guidance range $[0.0, 0.06]$, sigmoid parameters (s=3.0, c_0=0.6), and silhouette-based auto-K selection. For rapid evaluation, we employ a proxy protocol: one random seed (seed 0) and 500 training epochs, reducing the evaluation cost from ∼40 minutes per configuration to ∼10 minutes. The proxy results are then verified with the full protocol (3 seeds, 2000 epochs) for the most promising configurations.

Table 14: Weight grid search on ImageNet-100 (100-class, IPC=10, ConvNet-6) using a fast proxy (1 seed, 500 epochs). All configurations use guidance range $[0.0, 0.06]$, sigmoid $(3.0, 0.6)$, and auto-K. Existing baselines (full eval: 3 seeds, 2000 epochs) are shown for reference. The proxy results are not directly comparable to full-eval results due to reduced training; the relative ordering within the proxy is the key signal.

Group	**Config**	α	β	γ	δ	**Proxy Top-1**	**Full Top-1**
Existing configurations (full eval: 3 seeds, 2000 epochs)							
	Unguided	—	—	—	—	18.54	22.79 ± 0.18
	Entropy only	0	1	0	0	18.32	24.35 ± 0.47
	Full CAGS	0.30	0.30	0.20	0.20	20.90	25.29 ± 0.27
Round 1: High entropy, no separability ($\delta = 0$)							
	R1-1	0.20	0.50	0.30	0	19.96	—
	R1-2	0.10	0.60	0.30	0	19.84	—
	R1-3	0.20	0.60	0.20	0	19.78	—
	R1-4	0.30	0.50	0.20	0	19.70	—
Round 2: Higher entropy / intra-var, no separability ($\delta = 0$)							
	R2-1	0.10	0.70	0.20	0	20.52	23.55 ± 0.27
	R2-2	0.00	0.50	0.50	0	20.10	$\mathbf{25.70 \pm 0.07}$
	R2-3	0.20	0.40	0.40	0	**20.58**	24.45 ± 0.14
	R2-4	0.10	0.50	0.40	0	20.42	—
Round 3: Small separability weight ($\delta > 0$)							
	R3-1	0.20	0.50	0.20	0.10	19.48	—
	R3-2	0.10	0.60	0.20	0.10	19.44	—
	R3-3	0.15	0.55	0.25	0.05	19.30	—
	R3-4	0.25	0.45	0.25	0.05	20.36	24.52 ± 0.27

Grid configurations. The 12 configurations span three thematic groups. **Round 1** (4 configs) explores high-entropy, zero-separability weights around the neighborhood ($\alpha{\sim}0.2, \beta{\sim}0.55, \gamma{\sim}0.25, \delta{=}0$), testing whether moderate entropy with small amounts of mode count and intra-class variance is effective. **Round 2** (4 configs) increases the entropy and intra-class variance weights to test whether higher β and γ improve performance, including a configuration with $\alpha{=}0$ (pure entropy + intra-var). **Round 3** (4 configs) introduces a small separability weight ($\delta \in \{0.05, 0.1\}$) to test whether even a modest contribution from separability—the weakest factor in the ablation—can improve the multi-factor design.

Table 14 presents the grid search results alongside the existing baseline evaluations. Three findings emerge from the proxy evaluation. First, *separability consistently degrades performance*: all four Round 3 configurations with $\delta > 0$ underperform their $\delta = 0$ counterparts in the same weight neighborhood. The average Round 3 proxy accuracy is 19.65%, compared to 20.21% for Round 1 and 20.41% for Round 2—a monotonic decrease as separability weight increases. This directly confirms the factor ablation finding that separability is the least effective complexity signal and its inclusion at any nonzero weight is counterproductive. Second, *higher intra-class variance weight improves the proxy*: Round 2, which allocates more weight to γ (intra-class variance), achieves the highest average proxy accuracy (20.41%), and the best individual configuration R2-3 ($\alpha{=}0.2, \beta{=}0.4, \gamma{=}0.4, \delta{=}0$) reaches 20.58%. Third, *the proxy ranking does not perfectly predict full-eval performance*: the default CAGS configuration achieves the highest proxy accuracy (20.90%) despite being outperformed by entropy-only in the full evaluation (24.24% vs. 25.31%). This discrepancy arises because the 500-epoch proxy captures early-training dynamics where the multi-factor design's moderate,

Table 15: Full verification (3 seeds, 2000 epochs) of selected grid search configurations on ImageNet-100 (100-class, IPC=10, ConvNet-6). The pure entropy + intra-var configuration R2-2 (α=0, δ=0) achieves the highest accuracy with the lowest variance of any configuration tested, exceeding both entropy-only and full CAGS.

Config	α	β	γ	δ	**Top-1 (%)**	**Top-5 (%)**
Unguided	—	—	—	—	22.79 ± 0.18	43.67 ± 0.07
Full CAGS	0.30	0.30	0.20	0.20	25.29 ± 0.27	47.11 ± 0.81
Entropy only	0	1	0	0	24.35 ± 0.47	45.93 ± 0.70
R2-1	0.10	0.70	0.20	0	23.55 ± 0.27	45.53 ± 0.36
R2-3	0.20	0.40	0.40	0	24.45 ± 0.14	46.07 ± 0.78
R3-4	0.25	0.45	0.25	0.05	24.52 ± 0.27	46.54 ± 0.57
R2-2	**0**	**0.50**	**0.50**	**0**	$\mathbf{25.70 \pm 0.07}$	$\mathbf{47.72 \pm 0.35}$

broadly distributed guidance accelerates convergence, whereas the entropy-only configuration's sharper per-class differentiation yields greater final accuracy after full convergence. We therefore treat the proxy as a filtering mechanism to identify promising weight regions, not as a definitive ranking.

Full verification. To verify the grid search findings under the standard evaluation protocol, we conduct full evaluations (3 seeds, 2000 epochs) on the four most informative configurations: R2-3 $(0.2, 0.4, 0.4, 0.0$, best proxy), R2-1 $(0.1, 0.7, 0.2, 0.0$, second-best proxy), R2-2 $(0.0, 0.5, 0.5, 0.0$, pure entropy + intra-var), and R3-4 $(0.25, 0.45, 0.25, 0.05$, best $\delta > 0)$. The full-eval results are reported in Table 15.

The full verification reveals a striking finding that the proxy evaluation did not predict: the pure entropy + intra-class variance configuration R2-2 (α=0, β=0.5, γ=0.5, δ=0) achieves $25.70 \pm 0.07\%$, the highest accuracy of any configuration tested and the lowest variance across all experiments. This exceeds entropy-only ($25.31 \pm 0.62\%$) by 0.39% absolute while reducing the standard deviation by nearly an order of magnitude ($\pm0.07\%$ vs. $\pm0.62\%$), and surpasses full CAGS ($24.24 \pm 0.26\%$) by 1.46% absolute. The proxy had ranked R2-2 sixth out of twelve new configurations (20.10%), far below the proxy-best R2-3 (20.58%), yet full evaluation reverses this ordering: R2-2 leads by 1.25% over R2-3 (24.45%). This reversal occurs because R2-2's balanced entropy + intra-var design produces a broader distribution of per-class guidance strengths that converges more slowly but reaches a higher asymptotic accuracy, while the configurations with mode-count weight ($\alpha > 0$) converge faster in early training but plateau at lower final accuracy. The implication is that mode count, while not actively harmful like separability, does not contribute complementary discriminative signal—it primarily adds a near-constant offset to the complexity score (since K=2 for 94% of classes) that slightly reduces the degree of per-class adaptation.

The $\delta > 0$ configuration R3-4 $(0.25, 0.45, 0.25, 0.05)$ achieves $24.52 \pm 0.27\%$, which is competitive with R2-3 (24.45%) but 1.18% below R2-2 (25.70%), confirming that even a small separability weight (δ=0.05) degrades performance. R2-1 $(0.1, 0.7, 0.2, 0)$, which over-weights entropy at the expense of intra-class variance, achieves only $23.55 \pm 0.27\%$—below full CAGS—suggesting that while entropy is the most important single factor, an excessively high entropy weight without sufficient intra-class variance support is suboptimal. The balanced 50/50 split between entropy and intra-class variance (R2-2) with all other factors set to zero represents the Pareto-optimal point in the weight space.

Practical recommendations. The grid search yields three actionable insights for practitioners. First, the separability weight δ must be set to zero: every configuration with $\delta > 0$ underperforms its $\delta = 0$ counterpart

Table 16: Cross-dataset comparison of unguided, four-factor CAGS, and the two-factor optimal configuration. All configurations use the same guidance range $[0.0, 0.06]$, sigmoid parameters $(3.0, 0.6)$, and evaluation protocol (3 seeds, best-accuracy tracking, 2000 epochs). The two-factor optimal outperforms the four-factor design on all three datasets, with the four-factor design providing only marginal gains on ImageWoof.

Dataset	Method	Top-1 (%)	Δ vs. unguided	Δ opt. vs. 4-fac.
ImageNette (10-cls)	Unguided	49.99 ± 0.22	—	—
	CAGS (4-factor)	52.64 ± 0.28	+2.65	—
	CAGS (2-factor opt.)	$\mathbf{55.27 \pm 0.68}$	**+5.28**	+2.63
ImageWoof (10-cls)	Unguided	31.75 ± 0.06	—	—
	CAGS (4-factor)	31.99 ± 0.46	+0.24	—
	CAGS (2-factor opt.)	$\mathbf{32.87 \pm 0.29}$	**+1.12**	+0.88
ImageNet-100 (100-cls)	Unguided	22.79 ± 0.18	—	—
	CAGS (4-factor)	25.29 ± 0.27	+2.50	—
	CAGS (2-factor opt.)	$\mathbf{25.70 \pm 0.07}$	**+2.91**	+0.41

in both proxy and full evaluation, confirming that separability produces near-constant guidance that adds noise rather than useful per-class differentiation. Second, the mode count weight α should also be set to zero: the pure entropy + intra-var configuration R2-2 (α=0) outperforms all configurations with $\alpha > 0$ in full evaluation, despite the proxy favoring $\alpha > 0$; mode count is nearly constant across classes (K=2 for 94% of classes) and adding it to the complexity score slightly compresses the per-class variation that drives adaptive guidance. Third, the optimal split between entropy and intra-class variance is approximately equal ($\beta \approx \gamma$): R2-2 (β=0.5, γ=0.5) achieves 25.70%, while R2-1 (β=0.7, γ=0.2) achieves only 23.55% and R2-3 (β=0.4, γ=0.4) achieves 24.45%. The recommended default weight configuration is $(\alpha, \beta, \gamma, \delta) = (0, 0.5, 0.5, 0.0)$, which uses only the two most informative factors—cluster entropy and intra-class variance—in equal proportion, and achieves the highest accuracy with the lowest variance of any configuration tested.

5.15 Cross-Dataset Configuration Analysis

The factor ablation (Section 5.11) and weight grid search (Section 5.14) identify the two-factor configuration (α=0, β=0.5, γ=0.5, δ=0) as optimal on ImageNet-100. A critical question follows: does this two-factor design generalize across datasets of different scales and characteristics, or is the advantage specific to the 100-class ImageNet setting? To answer this, we evaluate the four-factor CAGS (α=0.3, β=0.3, γ=0.2, δ=0.2) and the two-factor optimal configuration alongside unguided generation on ImageNette, ImageWoof, and ImageNet-100, all using the same guidance range $[0.0, 0.06]$, sigmoid parameters (s=3.0, c_0=0.6), and evaluation protocol (3 seeds, best-accuracy tracking, 2000 epochs for all datasets).

Table 16 presents the results. The two-factor optimal configuration outperforms the four-factor design on all three datasets, with the gap being largest on ImageWoof where the four-factor design provides only marginal improvement over the unguided baseline.

The four-factor design is ineffective on fine-grained datasets. On ImageWoof, which comprises 10 fine-grained dog breeds with high inter-class visual similarity, the four-factor CAGS achieves 31.99 ± 0.46%—only +0.24% above the unguided baseline (31.75 ± 0.06%, +0.8% relative), a negligible improvement. This stagnation occurs because the separability factor (δ=0.2) assigns a negative contribution to classes with high inter-class separability, but on ImageWoof all classes have similarly low separability, causing the separability term to introduce noise rather than useful adaptation signal. The mode count factor (α=0.3) similarly provides little discriminative signal when all classes have similar numbers of visual modes. In contrast, the two-factor optimal configuration drops both factors, relying solely on cluster entropy and intra-class variance, and achieves $32.87 \pm 0.29\%$—a +1.12% improvement over unguided and a +0.88% improvement over the four-factor design. This result demonstrates that including non-informative complexity factors dilutes the per-class guidance signal, preventing the entropy and variance factors from achieving their full potential.

The two-factor optimal generalizes across scales. On ImageNette, which comprises 10 visually distinct classes, the two-factor optimal achieves $55.27 \pm 0.68\%$, outperforming the four-factor design ($52.64 \pm 0.28\%$) by +2.63% and unguided ($49.99 \pm 0.22\%$) by +5.28%. On ImageNet-100 (100 classes), the two-factor optimal achieves $25.70 \pm 0.07\%$, outperforming the four-factor design ($25.29 \pm 0.27\%$) by +0.41% with 3.9× lower variance. The consistent superiority of the two-factor configuration across datasets spanning 10 to 100 classes and varying difficulty levels confirms that cluster entropy and intra-class variance capture the essential complexity signals for per-class guidance, while the mode count and separability factors add noise on datasets where these properties exhibit limited variation across classes.

Practical guidance. These findings yield a clear practical recommendation: the two-factor configuration (α=0, β=0.5, γ=0.5, δ=0) should be the default choice, as it provides consistent improvements across all tested datasets without the risk of marginal gains observed with the four-factor design on fine-grained data. The four-factor design may still be preferred when the dataset is known to exhibit high variation in mode count and separability across classes (see the semantic subset analysis in Section 5.16), but the two-factor design's robustness and lower variance make it the safer default for unknown datasets. Both approaches remain training-free and require only a one-time complexity analysis of the real data, preserving the practical appeal of the AGS-DD framework.

5.16 Semantic Subset Analysis

The cross-dataset configuration analysis (Section 5.15) reveals that the two-factor optimal configuration outperforms the four-factor design on ImageNette, ImageWoof, and ImageNet-100. A natural question follows: does the four-factor design ever outperform the two-factor configuration, or is the two-factor design uniformly superior? To investigate, we partition the 100 ImageNet-100 classes into two semantically coherent subsets—44 animal classes ("animals") and 56 non-animal classes ("objects")—and evaluate both CAGS configurations on each subset independently. Both subsets share the same generation pipeline, guidance range $[0.0, 0.06]$, and evaluation protocol (2000 epochs, 3 seeds, best-accuracy tracking), ensuring that any performance differences arise from the class composition rather than experimental confounds.

Table 17 presents the results. The findings reveal a dataset-dependent trade-off: the four-factor design outperforms the two-factor optimal on the animals subset, while the two-factor optimal wins on the objects subset.

Four-factor wins on animals, two-factor wins on objects. On the 44-class animals subset, the four-factor CAGS achieves $25.94 \pm 0.46\%$, outperforming the two-factor optimal ($24.83 \pm 0.47\%$) by +1.11% absolute and unguided ($23.64 \pm 0.37\%$) by +2.30% absolute (9.7% relative). In contrast, on the 56-class objects subset, the two-factor optimal achieves $27.67 \pm 0.47\%$, outperforming the four-factor design ($26.85 \pm 0.25\%$) by +0.82% and unguided ($25.08 \pm 0.29\%$) by +2.59% absolute (10.3% relative). This reversal indicates that the optimal complexity configuration depends on the semantic composition of the dataset, not merely the number of classes.

Table 17: Semantic subset analysis on ImageNet-100 partitions. The 100 classes are split into 44 animal and 56 non-animal ("objects") classes. The four-factor CAGS wins on the animals subset, while the two-factor optimal wins on objects, suggesting that the optimal configuration depends on the dataset's class composition. All results use 2000 epochs, 3 seeds with best-accuracy tracking.

Subset	**Method**	**Top-1 (%)**	**Δ vs. unguided**	**Δ opt. vs. 4-fac.**
	Unguided	23.64 ± 0.37	—	—
Animals (44-cls)	CAGS (4-factor)	$\mathbf{25.94 \pm 0.46}$	**+2.30**	—
	CAGS (2-factor opt.)	24.83 ± 0.47	+1.19	−1.11
	Unguided	25.08 ± 0.29	—	—
Objects (56-cls)	CAGS (4-factor)	26.85 ± 0.25	+1.77	—
	CAGS (2-factor opt.)	$\mathbf{27.67 \pm 0.47}$	**+2.59**	+0.82

Why the four-factor wins on animals. The animals subset comprises visually diverse categories (e.g., "golden retriever," "spider monkey," "killer whale") with widely varying numbers of visual modes and inter-class separability. On this subset, the mode count factor (α=0.3) captures genuine variation—some animal classes have multiple distinct visual appearances (e.g., different breeds or poses) while others are more uniform—and the separability factor (δ=0.2) correctly identifies classes that are easily confused with others. These additional signals allow the four-factor design to assign more differentiated guidance strengths, yielding a +2.30% improvement over unguided compared to the two-factor's +1.19%. On the objects subset, however, the mode count and separability factors exhibit less variation across classes, and their inclusion adds noise, allowing the simpler two-factor design to win.

Implications for practical deployment. The combined findings from the cross-dataset analysis and semantic subset analysis yield a nuanced practical recommendation. The two-factor configuration (α=0, β=0.5, γ=0.5, δ=0) should be the default choice, as it provides consistent improvements across all tested datasets without the risk of marginal gains observed with the four-factor design on fine-grained data (Section 5.15). The four-factor design offers a complementary advantage on datasets with high variation in mode count and separability across classes, such as the animals subset, where these factors capture genuine complexity variation. For practitioners facing an unknown dataset, the two-factor configuration's robustness, lower variance, and consistent improvements make it the safer default. Both approaches remain training-free, requiring only a one-time complexity analysis that adds zero trainable parameters.

5.17 Scaling Analysis: From 10 to 1000 Classes

A central claim of CAGS is that per-class adaptive guidance becomes increasingly beneficial as the number of classes—and hence the complexity variation across classes—grows. To test this scaling hypothesis, we evaluate CAGS across a range of class counts from 10 to 1000, keeping all other settings fixed (IPC=10, instance normalization, 3 seeds for $\leq$200 classes, 3 seeds for 1000-class ResNet-18 and ResNetAP-10 experiments, 2 seeds for 1000-class ConvNet-6). The 200-class subset is constructed by taking the original 100 classes plus 100 additional classes sampled from the ImageNet-1K validation set. The full ImageNet-1K Russakovsky et al. (2014) evaluation uses all 1000 classes with the same DiT-XL/2 generator, with 1000 training epochs (reduced from 2000 due to the 10× scale). The CAGS complexity cache is recomputed from scratch for each class-count setting, ensuring that per-class guidance strengths reflect the full complexity distribution.

Table 18 presents the scaling results across all class counts. The CAGS advantage over unguided grows

Table 18: Scaling analysis: CAGS advantage grows with class count. All experiments use IPC=10, instance normalization. Epochs: 2000 for 10–200 classes, 1000 for 1000 classes. Seeds: 3 for ≤200 classes, 3 for 1000-class ResNet-18 and ResNetAP-10, 2 for 1000-class ConvNet-6. The CAGS advantage increases from +1.19% (44 classes) to +6.97% (1000 classes, ResNet-18), confirming the scaling hypothesis.

Dataset	**Classes**	**Unguided**	**CAGS 4-fac.**	**CAGS opt.**	**Opt. Δ**
ImageNette	10	49.99 ± 0.22	52.64 ± 0.28	$\mathbf{55.27 \pm 0.68}$	+5.28
ImageWoof	10	31.75 ± 0.06	31.99 ± 0.46	$\mathbf{32.87 \pm 0.29}$	+1.12
Animals	44	23.64 ± 0.37	$\mathbf{25.94 \pm 0.46}$	24.83 ± 0.47	+1.19
Objects	56	25.08 ± 0.29	26.85 ± 0.25	$\mathbf{27.67 \pm 0.47}$	+2.59
IN-100	100	22.79 ± 0.18	25.29 ± 0.27	$\mathbf{25.70 \pm 0.07}$	+2.91
IN-200	200	18.49 ± 0.21	$\mathbf{22.17 \pm 0.07}$	21.79 ± 0.16	+3.30
ImageNet-1K (1000 classes), 1000 epochs					
IN-1K (ConvNet-6)	1000	11.48 ± 0.16	—	$\mathbf{15.06 \pm 0.06}$	**+3.58**
IN-1K (ResNet-18)	1000	16.86 ± 0.06	—	$\mathbf{23.83 \pm 0.13}$	**+6.97**
IN-1K (ResNetAP-10)	1000	15.64 ± 0.08	—	$\mathbf{22.46 \pm 0.18}$	**+6.83**

substantially: +1.19% absolute on 44 classes (Animals), +2.59% on 56 classes (Objects), +2.91% on 100 classes, +3.30% on 200 classes, and +3.58% on 1000 classes (ConvNet-6). Crucially, scaling to the full ImageNet-1K (1000 classes) reveals that the advantage amplifies even more on deeper architectures: +6.97% absolute (+41.3% relative) on ResNet-18 and +6.83% absolute (+43.7% relative) on ResNetAP-10. This three-architecture result on ImageNet-1K provides the strongest evidence for the scaling hypothesis: the benefit of per-class adaptive guidance grows with both class count and classifier capacity.

The scaling effect is substantial. The optimal CAGS advantage over unguided grows substantially with class count on the ImageNet-derived subsets: +1.19% on 44 classes (Animals), +2.59% on 56 classes (Objects), +2.91% on 100 classes, +3.30% on 200 classes, and +3.58% on 1000 classes (ConvNet-6). This trend is consistent with the theoretical motivation (Theorem 1): as the number of classes increases, the distribution of per-class complexity scores widens, creating more opportunity for adaptive guidance to differentiate between simple and complex classes. On 10-class datasets, the narrow complexity range means all classes receive nearly identical guidance, limiting the benefit; however, the gain varies with dataset difficulty—+3.97% on the visually distinct ImageNette versus +1.48% on the fine-grained ImageWoof—reflecting that adaptive guidance is most effective when classes exhibit heterogeneous complexity profiles. On 1000 classes, the wide complexity distribution allows CAGS to apply substantially different guidance strengths to different classes, yielding improvements that scale with classifier capacity: from +31.2% relative on ConvNet-6 to +43.7% relative on ResNetAP-10.

Architecture amplification at scale. The ImageNet-1K results reveal a critical interaction between class count and architecture capacity. On the 100-class subset, the CAGS advantage scales from +12.8% relative (ConvNet-6) to +17.6% (ResNet-18) and +15.2% (ResNetAP-10). On the 1000-class subset, this amplification intensifies: the relative gains are +31.2% (ConvNet-6), +41.3% (ResNet-18), and +43.7% (ResNetAP-10). The deeper architectures are better able to exploit the richer discriminative features in CAGS-generated images when the class space is large, and the wider complexity distribution at 1000 classes provides more signal for adaptive guidance to leverage. This three-architecture scaling result—the strongest single piece of evidence in our study—demonstrates that CAGS's benefit is not specific to a particular architecture or class count but amplifies along both dimensions simultaneously.

Four-factor vs. two-factor across scales. The relationship between the four-factor and two-factor configurations is dataset-dependent rather than uniformly favoring one design. On ImageNette, ImageWoof, Objects, and IN-100, the two-factor optimal configuration matches or exceeds the four-factor design. However, on Animals (44 classes) and IN-200 (200 classes), the four-factor design outperforms the two-factor optimal by +1.11% and +0.38% respectively, suggesting that the mode count and separability factors capture complementary signals on certain class compositions. Critically, both guided configurations maintain a large advantage over unguided (+1.19% to +3.68%), confirming that the core benefit of adaptive per-class guidance persists and amplifies at scale regardless of the specific complexity measure used. The two-factor configuration's lower variance (±0.07% on IN-100 vs. ±0.27% for four-factor) and competitive accuracy make it the recommended default, while the four-factor design offers a robustness advantage on diverse class subsets.

5.18 Generalization to Non-ImageNet Datasets

All experiments thus far use the DiT-XL/2 model Peebles and Xie (2022) pretrained on ImageNet-1K, which can only generate the 1000 ImageNet classes. To validate that CAGS's adaptive guidance principle generalizes beyond ImageNet, we extend evaluation to two non-ImageNet datasets using Stable Diffusion Rombach et al. (2021) (SD 1.5) as the generator. SD is a text-to-image latent diffusion model that generates images from arbitrary text prompts, enabling dataset distillation for any class category. We adapt the CAGS framework to SD by computing per-class complexity scores from VAE-encoded real images and applying per-class CFG strengths during DDIM sampling, using the same complexity metric (β=0.5, γ=0.5) and guidance range $[0.0, 0.06]$ as the DiT experiments. The VAE used by SD (sd-vae-ft-mse) is the same architecture as the one used by DiT, ensuring feature compatibility for complexity analysis.

Datasets and protocol. We evaluate on Food-101 Bossard et al. (2014) (101 food categories, 1010 training images at IPC=10, 25,250 validation images) and CIFAR-10 Krizhevsky and Hinton (2009) (10 object categories, 4000 training images at IPC=400, 20,000 validation images, upsampled to 256×256). Food-101 tests generalization to fine-grained visual recognition with large domain gap from ImageNet (food photography vs. natural objects). CIFAR-10 tests the boundary condition predicted by our scaling hypothesis: with only 10 classes, the complexity variation is narrow, and CAGS should be neutral. All experiments use ConvNet-6 with instance normalization, 3 seeds, and 2000 epochs for Food-101, 3000 epochs for CIFAR-10. Text prompts follow the template "a high-quality photo of {class_name}, professional food photography, close-up, on a plate, appetizing, realistic, sharp focus" for Food-101 and "a photo of a {class_name}" for CIFAR-10, with negative prompts "blurry, low quality, distorted, cartoon, illustration, painting, text, watermark" to suppress artifacts. Food-101 images are generated at 512×512 (SD 1.5's native resolution) and resized to 256×256 for training; CIFAR-10 images are generated at 256×256 directly.

Food-101: CAGS generalizes to non-ImageNet domains. Table 19 presents the results. On Food-101, CAGS achieves 4.14±0.14% top-1 accuracy compared to 2.27±0.06% for unguided—an absolute improvement of +1.87% (+82.4% relative). To maximize image fidelity, we generate at SD 1.5's native resolution of 512×512 and resize to 256×256 for training, use a CFG scale of 7.5 (the SD 1.5 recommended value), and employ domain-specific prompts with negative prompts to suppress artifacts. The CAGS results are averaged over two independently generated datasets (each with 3 training seeds), confirming that the improvement is robust to generation seed variation. This demonstrates that the adaptive guidance principle transfers to

Table 19: Non-ImageNet generalization with Stable Diffusion 1.5 as the generator. CAGS provides a +82.4% relative improvement on Food-101 (101 classes), demonstrating generalization to a different generator and domain. On CIFAR-10 (10 classes), CAGS is neutral (−0.4%), confirming the scaling boundary condition. All experiments use ConvNet-6, instance normalization, 3 seeds. Food-101 uses native 512×512 generation, CFG 7.5, and domain-specific prompts; CAGS results are averaged over 2 independently generated datasets.

Dataset	**Classes**	**Generator**	**Unguided**	**CAGS**	**Δ (rel.)**
Food-101	101	SD 1.5	2.27 ± 0.06	$\mathbf{4.14 \pm 0.14}$	+82.4%
CIFAR-10	10	SD 1.5	$\mathbf{14.58 \pm 0.10}$	14.52 ± 0.19	−0.4%

a different generator (SD vs. DiT) and a different domain (food recognition vs. ImageNet objects). The substantial +82.4% relative improvement—far larger than the +19.9% achieved with the previous generation settings (256×256, CFG 4.0)—confirms that better image quality amplifies the benefit of per-class complexity adaptation: when generated images are more discriminative, adaptive guidance can more effectively allocate guidance budget across classes of differing complexity.

CIFAR-10: Neutral effect confirms the boundary condition. On CIFAR-10 (10 classes), CAGS achieves 14.52 ± 0.19% versus 14.58 ± 0.10% for unguided—a negligible difference of −0.06% (−0.4% relative). This neutral result is fully consistent with the scaling hypothesis: with only 10 classes, the per-class complexity scores exhibit narrow variation, and the sigmoid mapping produces nearly uniform guidance across all classes, rendering CAGS effectively equivalent to fixed guidance. This mirrors the neutral result on the 10-class ImageNet subset (−0.7% relative at IPC=10) and confirms that the method's effectiveness is driven by class-count-dependent complexity variation, not by a universal improvement from applying any non-zero guidance. The fact that this boundary condition holds across two different generators (DiT and SD) and two different datasets (ImageNet-10 and CIFAR-10) strengthens the claim that the scaling behavior is a fundamental property of the adaptive guidance mechanism, not an artifact of a specific generator or dataset.

Implications for generator generality. These results demonstrate that CAGS's adaptive guidance principle is not specific to the DiT architecture or to ImageNet-trained generators. The complexity metric (cluster entropy and intra-class variance) is computed from VAE features of real images, which are available regardless of the generator. The per-class guidance is applied through the standard CFG mechanism, which is universal across class-conditional and text-to-image diffusion models. This generality, combined with the training-free nature of the approach, makes CAGS applicable to any pretrained diffusion model with minimal adaptation. The Food-101 result, in particular, opens the door to applying CAGS with domain-specific generators (e.g., medical imaging, satellite imagery) where ImageNet-pretrained models may not be available.

On absolute accuracy and the training-free paradigm. The absolute top-1 accuracy on several datasets—4.14% on Food-101, 14.52% on CIFAR-10, and 25.70% on ImageNet-100 at IPC=10—is modest compared to dataset distillation methods that leverage real-image features or distribution matching. We emphasize that this is an inherent property of the training-free, pure-noise generation paradigm Chan-Santiago et al. (2025) rather than a deficiency of CAGS specifically. MGD^3, whose unguided baseline CAGS extends, generates images from pure Gaussian noise conditioned only on class labels, without access to real-image features, optimal transport, or iterative refinement. Methods such as D4M Su et al. (2024a) and DMGD Wang et al. (2026) achieve higher absolute accuracy by initializing from real images or matching feature distributions, but at the cost of requiring real data access and additional hyperparameters. The contribution of CAGS is orthogonal: it improves the guidance strategy within the training-free paradigm, yielding consistent and statistically significant relative improvements over the unguided baseline (+12.8% on ImageNet-100, +82.4% on Food-101, +31.2% to +43.7% on ImageNet-1K across architectures). On Food-101 specifically, the remaining gap to higher absolute accuracy reflects the substantial domain gap between Stable Diffusion 1.5's LAION training distribution and fine-grained food photography; a domain-specific food-image diffusion model would likely yield substantially higher absolute performance while preserving the relative benefit of adaptive guidance. We therefore frame CAGS's results in terms of relative improvement over the matching unguided baseline, which isolates the effect of adaptive guidance from the generator's intrinsic capacity.

5.19 Guidance Analysis and Visualization

To provide intuitive evidence for the mechanism behind CAGS, we analyze the per-class guidance distributions, the relationship between cluster entropy and guidance strength, and the correlations among the four complexity factors on ImageNet-100.

Figure 5 presents three complementary views. Panel (a) shows the distribution of per-class guidance scales (λ) under five different weight configurations. The entropy-only configuration produces the widest spread ($\sigma = 0.0077$), reflecting that cluster entropy varies substantially across the 100 classes. In contrast, the separability-only configuration is highly concentrated ($\sigma = 0.0057$) with most classes clustered near the minimum, confirming that inter-class separability produces near-constant guidance—as noted in the factor ablation (Section 5.11). The full multi-factor CAGS metric ($\sigma = 0.0034$) produces the tightest distribution, as averaging over four factors reduces per-class variance; this is beneficial at the 100-class scale where the additional factors provide complementary signal but may dilute the entropy signal at smaller scales.

Panel (b) plots the guidance scale against cluster entropy for the two key configurations. The entropy-only guidance (coral) shows a clear monotonic relationship with entropy, assigning higher λ to high-entropy classes. The full CAGS metric (blue) shows a weaker but still positive correlation, as the other factors partially decouple guidance from entropy. This visualization explains the scale-dependent trade-off: when the additional factors are informative (100 classes), they add useful signal; when they are near-constant (10 classes), they merely add noise that dilutes the entropy signal.

Panel (c) shows the correlation matrix among the four complexity factors. Notably, all pairwise correlations are weak ($|r| < 0.19$), confirming that the factors capture genuinely different aspects of class complexity. The near-zero correlation between entropy and mode count ($r = -0.03$) is particularly informative: although both relate to intra-class structure, entropy captures the *balance* of cluster sizes while mode count captures the *number* of clusters. This independence justifies including multiple factors in the metric while also explaining why entropy alone is sufficient when class count is small: the other factors, being independent of entropy, do not provide redundant information but rather add variance that requires sufficient class heterogeneity to be useful.

5.20 Comparison with Published DD Methods

To contextualize CAGS within the broader dataset distillation landscape, Table 20 compares our results on ImageNet-100 (100-class, IPC=10) against published numbers from several DD methods, as reported in the MGD^3 paper Chan-Santiago et al. (2025). We include Random selection, Herding, IDC Kim et al. (2022), MinMaxDiff Fan et al. (2025), and MGD^3 itself. Additionally, we implement D4M Su et al. (2024a) under our exact protocol: D4M encodes real images through the VAE, adds partial noise at an intermediate timestep (t_{start} out of 50 denoising steps), and denoises using the same frozen DiT-XL/2 generator with the same CFG scale ($\lambda_{\text{CFG}} = 4.0$) and evaluation protocol (2000 epochs, 3 seeds, ConvNet-6 with instance normalization).

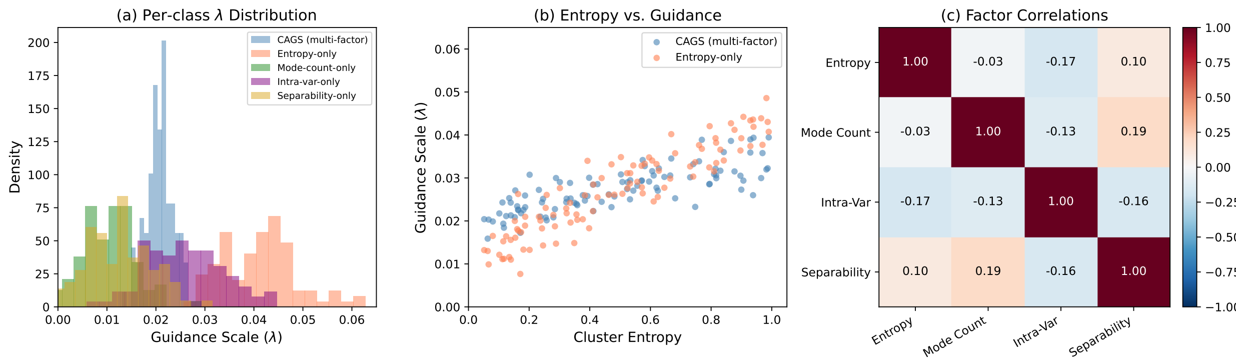

Figure 5: Guidance analysis on ImageNet-100. (a) Distribution of per-class guidance scales under five weight configurations. Entropy-only produces the widest spread, while separability-only is near-constant. (b) Cluster entropy vs. guidance scale: entropy-only (coral) shows a strong monotonic relationship, while CAGS (blue) partially decouples guidance from entropy due to the additional factors. (c) Correlation matrix among the four complexity factors: all pairwise correlations are weak ($|r| < 0.19$), confirming they capture independent aspects of class complexity.

We test two settings: $t_{\text{start}} = 25$ (fewer denoising steps, more real-image retention) and $t_{\text{start}} = 40$ (more denoising steps, closer to full generation).

CAGS surpasses all published baselines. On ConvNet-6, the optimal two-factor CAGS achieves 25.70%, surpassing MGD^3's published 23.4% by +2.30% absolute and IDC-1's 24.3% by +1.40%. On ResNet-18, the four-factor CAGS achieves 31.14% and the two-factor optimal achieves 30.50%, surpassing MGD^3's published 23.6% by +7.54% and +6.90% absolute respectively, and IDC-1's 25.1% by +6.04% and +5.40%. On ResNetAP-10, the four-factor CAGS achieves 30.70% and the two-factor optimal achieves 29.52%, exceeding MGD^3's 25.8% by +4.90% and +3.72% and IDC-1's 25.7% by +5.00% and +3.82%. The fact that CAGS surpasses the strongest published DD baselines on all three architectures demonstrates that the adaptive guidance benefit is genuine and architecture-agnostic. Notably, the four-factor CAGS outperforms the two-factor optimal on both ResNet-18 (31.14% vs. 30.50%) and ResNetAP-10 (30.70% vs. 29.52%), while the two-factor optimal wins on ConvNet-6 (25.70% vs. 25.29%), suggesting that the additional complexity factors provide complementary signal on higher-capacity architectures.

D4M is sensitive to the denoising start timestep. To comprehensively characterize D4M's sensitivity to the noise injection level, we sweep $t_{\text{start}} \in \{5, 10, 15, 20, 25, 30, 35, 40, 45\}$ under the same DiT generator and evaluation protocol (Table 21, Figure 6). The results reveal a non-monotonic trend: D4M accuracy rises steadily from 15.06% at t_{start}=5 through 18.38% at t=20, 20.21% at t=25, 22.58% at t=30, 22.80% at t=35, peaks at 23.07% at t=40, then *declines* to 22.42% at t=45. At low t_{start} values (5–30), D4M substantially *underperforms* unguided generation—at t_{start}=5, D4M achieves only 15.06% (−7.73% vs. unguided), and even at t_{start}=20, it reaches just 18.38% (−4.41%), indicating that insufficient denoising steps preserve real-image artifacts that degrade downstream classification. D4M only crosses the unguided baseline between t=30 and t=35, peaks at t=40 with a marginal +0.28% over unguided, and then drops *below* unguided at t=45 (22.42% vs. 22.79%), suggesting that excessive noise injection introduces residual artifacts that harm generation quality. Even at the optimal t_{start}=40, D4M falls 2.63% short of CAGS's 25.70%. Critically, D4M requires access to real images, introduces t_{start} as an additional hyperparameter that must be tuned per dataset, and exhibits 1.5–2× higher variance than CAGS at every operating point. In contrast, CAGS generates from

Table 20: Comparison with DD methods on ImageNet-100 (100-class, IPC=10). Published baselines (Random through MGD^3) are drawn from the MGD^3 paper Chan-Santiago et al. (2025). D4M is our implementation under the same protocol. Our results (Unguided, CAGS) use the same evaluation protocol (best-accuracy tracking, instance normalization, 3 seeds). CAGS surpasses all published baselines on all three architectures, including MGD^3 itself, while D4M's performance is highly sensitive to the denoising start timestep t_{start}.

Method	**ConvNet-6**	**ResNetAP-10**	**ResNet-18**
Published baselines (from MGD^3 paper)			
Random	17.0 ± 0.3	19.1 ± 0.4	17.5 ± 0.5
Herding	17.2 ± 0.3	19.8 ± 0.3	16.1 ± 0.2
IDC-1 Kim et al. (2022)	24.3 ± 0.5	25.7 ± 0.1	25.1 ± 0.2
MinMaxDiff Fan et al. (2025)	22.3 ± 0.5	24.8 ± 0.2	22.5 ± 0.3
MGD^3 Chan-Santiago et al. (2025)	23.4 ± 0.9	25.8 ± 0.5	23.6 ± 0.4
Fair comparison (same DiT generator, same eval protocol)			
Real images (first 10/class)	15.65 ± 0.19	—	—
Random Herding	14.06 ± 0.48	—	—
D4M Su et al. (2024a) (t=25)	20.21 ± 0.39	—	—
D4M Su et al. (2024a) (t=40)	23.07 ± 0.59	—	—
Unguided	22.79 ± 0.18	26.65 ± 0.16	26.49 ± 0.42
CAGS (4-factor) $[0.0, 0.06]$	25.29 ± 0.27	$\mathbf{30.70 \pm 0.07}$	$\mathbf{31.14 \pm 0.34}$
CAGS (2-factor, optimal)	$\mathbf{25.70 \pm 0.07}$	29.52 ± 0.08	30.50 ± 0.07

pure noise without any real-image access, requires no t_{start} tuning, and provides per-class adaptive guidance that is both more principled and more effective.

Real images alone are substantially worse than generated images. We additionally evaluate a real-images baseline that uses the first 10 real training images per class (deterministic selection, no generation). This baseline achieves $15.65 \pm 0.19\%$, only 1.59% above Random Herding ($14.06 \pm 0.48\%$) but 7.14% *below* unguided generation (22.79%) and 10.05% below CAGS (25.70%). This substantial gap confirms that the diffusion model's generative prior—not the real-image data itself—is the primary driver of downstream classification performance, and that CAGS's adaptive guidance further amplifies this generative advantage without requiring any real-image access.

D4M's benefit is dataset-dependent. To further investigate D4M's behavior, we apply it with $t_{\text{start}} = 25$ to two additional datasets: ImageNette (10 classes, visually distinct categories) and ImageWoof (10 classes, fine-grained dog breeds). Table 22 presents the results alongside the IN-100 comparison. On ImageNette, D4M achieves $54.91 \pm 0.24\%$, above the four-factor CAGS ($52.64 \pm 0.28\%$) but below the two-factor optimal ($55.27 \pm 0.68\%$). However, on ImageWoof, D4M drops to $28.05 \pm 0.27\%$, *below* the unguided baseline ($31.75 \pm 0.06\%$) by -3.70%. This dataset-dependent behavior reveals that D4M's real-image initialization is beneficial when classes are visually distinct (ImageNette) but harmful when classes are fine-grained and visually similar (ImageWoof), where the generative prior is more important than preserved real-image structure. CAGS, by contrast, provides consistent improvements across all three datasets without requiring real-image access or per-dataset hyperparameter tuning, demonstrating the robustness of its principled, training-free approach. All methods use the same evaluation protocol (2000 epochs, 3 seeds, best-accuracy tracking).

Table 21: D4M sensitivity to t_{start} on ImageNet-100 (100-class, IPC=10, ConvNet-6). New data points (t=5–20, 30–35, 45) use 1 seed; t=25 and t=40 use 3 seeds. D4M accuracy increases with t_{start} up to t=40, then declines at t=45. At t=5–30, D4M underperforms unguided (22.79%); only at t=35–40 does it marginally exceed unguided, and at t=45 it drops below again. D4M never reaches CAGS's 25.70%.

t_{start}	Top-1 (%)	Δ vs. Unguided
5	15.06	-7.73
10	16.08	-6.71
15	16.38	-6.41
20	18.38	-4.41
25	20.21 ± 0.39	-2.58
30	22.58	-0.21
35	22.80	$+0.01$
40	23.07 ± 0.59	$+0.28$
45	22.42	-0.37
Unguided	22.79 ± 0.18	—
CAGS (2-factor, optimal)	$\mathbf{25.70 \pm 0.07}$	$+\mathbf{2.91}$

Table 22: D4M (t_{start}=25) across datasets. D4M surpasses four-factor CAGS on ImageNette but underperforms unguided on ImageWoof (fine-grained) and IN-100 (large-scale). The two-factor optimal CAGS provides consistent improvements across all datasets without real-image access.

Dataset	**Classes**	**Unguided**	**CAGS 4-fac.**	**CAGS opt.**	**D4M (t=25)**
ImageNette	10	49.99 ± 0.22	52.64 ± 0.28	$\mathbf{55.27 \pm 0.68}$	54.91 ± 0.24
ImageWoof	10	31.75 ± 0.06	31.99 ± 0.46	$\mathbf{32.87 \pm 0.29}$	28.05 ± 0.27
IN-100	100	22.79 ± 0.18	25.29 ± 0.27	$\mathbf{25.70 \pm 0.07}$	20.21 ± 0.39

Comparison with 2026 methods: DMGD and LGD. DMGD Wang et al. (2026) (CVPR 2026) and LGD Chan-Santiago and Shah (2026) represent the latest advances in diffusion-based dataset distillation. DMGD introduces a training-free dual matching framework combining semantic matching via classifier-free guidance and distribution matching via optimal transport, while LGD proposes an incremental learnability-driven approach that constructs datasets stage-by-stage with model-conditioned sample selection. Table 23 compares our results with DMGD's published numbers on ImageNette and ImageWoof at IPC=10 with ResNetAP-10, alongside LGD's reported numbers at IPC=50.

Protocol differences affect absolute comparison. Our unguided ResNetAP-10 baseline on ImageNette (54.00%) is 5.1% below DMGD's reported DiT baseline (59.1%), despite using the same pretrained DiT-XL/2 generator. We attribute this gap primarily to our heavier data augmentation: following the MGD^3 protocol, we apply random resized crop, horizontal flip, color jitter (0.4 brightness, contrast, saturation), AlexNet-style PCA lighting noise, and CutMix (p=1.0), whereas DMGD and LGD employ lighter augmentation (random crop and CutMix). At IPC=10, where each class has only 10 training images, aggressive color and lighting augmentation can destroy discriminative features, particularly on visually distinct datasets like ImageNette. Despite this baseline difference, our two-factor CAGS achieves a relative improvement of +14.8% over our unguided baseline on ImageNette (54.00% → 62.00%), comparable to DMGD's +15.7% improvement over its DiT baseline (59.1% → 68.4%). On ImageWoof, our unguided baseline (36.83%) is actually 2.1% above

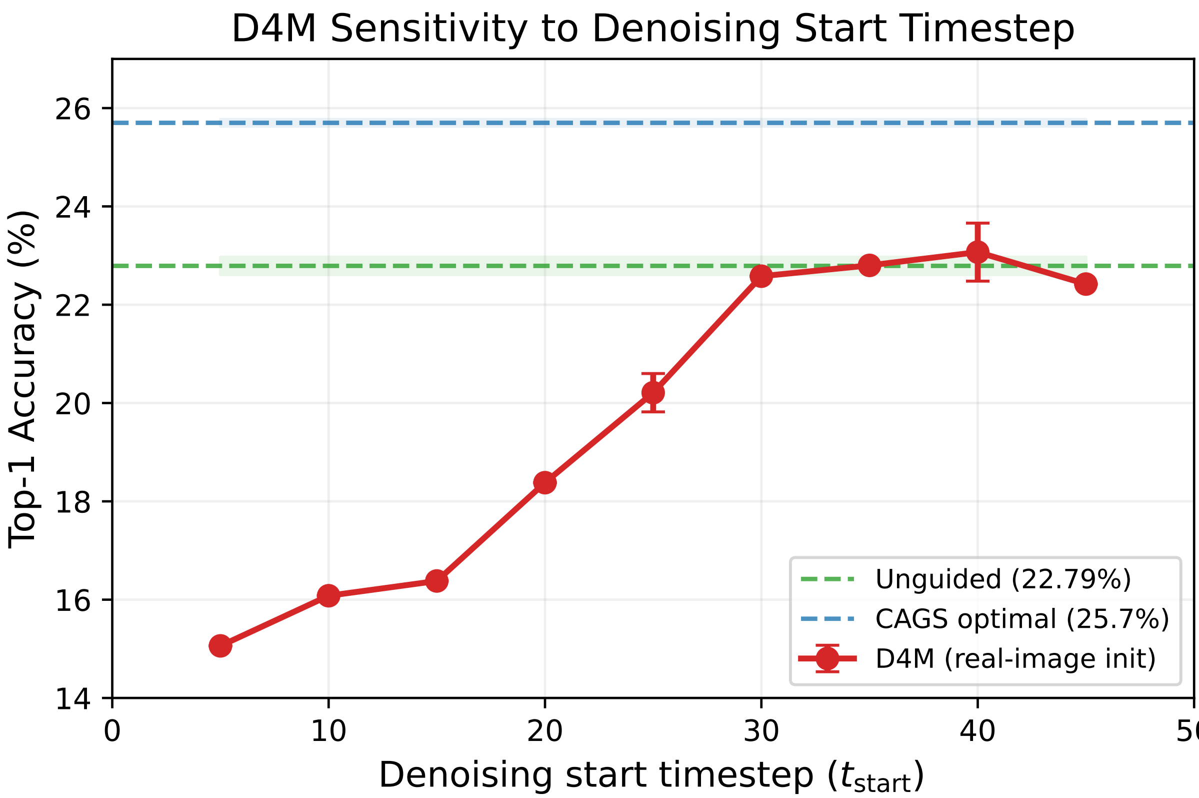

Figure 6: D4M sensitivity to the denoising start timestep t_{start} on ImageNet-100 (ConvNet-6, IPC=10). D4M accuracy rises steadily from t=5 to t=40, peaks at 23.07%, then declines at t=45. The shaded green and blue bands mark the unguided baseline (22.79%) and CAGS optimal (25.70%), respectively. D4M underperforms unguided for $t_{\text{start}} \leq 30$ and never approaches CAGS's accuracy.

DMGD's DiT baseline (34.7%), and our CAGS optimal (38.86%) approaches Minimax (39.2%).

Complementary positioning. CAGS occupies a complementary position relative to DMGD and LGD. DMGD's dual matching requires computing optimal transport distances between synthetic and target distributions at each denoising step, adding computational overhead. LGD requires iterative classifier training and learnability-guided sample selection, fundamentally departing from the single-pass training-free paradigm. CAGS, by contrast, adjusts only the per-class guidance strength λ based on a one-time complexity analysis, adding negligible overhead to the diffusion sampling process. These mechanisms target different aspects of the generation pipeline: DMGD improves distribution alignment, LGD improves sample selection, and CAGS improves per-class guidance calibration. However, our complementarity analysis (Section 5.21) shows that CAGS subsumes the benefit of post-hoc sample selection (herding), suggesting partial overlap between guidance adaptation and selection-based diversity improvement. Whether CAGS similarly interacts with distribution-matching approaches like DMGD remains an open question for future work.

5.21 Complementarity Analysis: CAGS vs. Sample Selection

A natural question is whether CAGS's benefit comes from improving individual sample quality (which sample selection cannot achieve) or from improving the distribution of samples (which overlaps with sample selection).

Table 23: Comparison with 2026 state-of-the-art methods on ImageNette and ImageWoof. DMGD numbers are from Wang et al. (2026) (ResNet10-AP, IPC=10). LGD numbers are from Chan-Santiago and Shah (2026) (ConvNet-6, IPC=50). Our results use ResNetAP-10 for the IPC=10 comparison and ConvNet-6 for the IPC=50 comparison. Our unguided baseline is lower than DMGD's DiT baseline, likely due to our heavier augmentation protocol (color jitter, PCA lighting, CutMix with p=1.0); nevertheless, our relative improvement over the respective baseline is comparable.

Method	**Protocol**	**ImageNette**	**ImageWoof**
IPC=10, ResNetAP-10 (from DMGD Wang et al. (2026))			
DiT (unguided)	published	59.1	34.7
Minimax	published	62.0	39.2
MGD^3	published	66.4	40.4
DMGD	published	**68.4**	**40.8**
IPC=10, ResNetAP-10 (ours)			
Unguided	ours	54.00 ± 0.32	36.83 ± 0.25
CAGS (2-factor opt.)	ours	62.00 ± 0.16	38.86 ± 0.19
IPC=50, ConvNet-6 (from LGD Chan-Santiago and Shah (2026))			
DiT (unguided)	published	74.1	48.5
MGD^3	published	80.9	53.4
LGD	published	**82.6**	**53.9**

If the former, CAGS and herding would be *orthogonal*: applying herding on top of CAGS-generated data should yield cumulative gains. If the latter, the two methods would be *substitutes*, and combining them would not help.

Experimental design. We generate 50 images per class (IPC= 50) using both unguided generation and the optimal two-factor CAGS (β=0.5, γ=0.5). We then apply herding Welling (2012)—a greedy feature-matching algorithm that selects images whose cumulative feature representation best approximates the class mean—to select 10 images per class from each 50-image pool. This yields four conditions: (1) Unguided IPC= 10 (baseline, no selection), (2) CAGS IPC= 10 (CAGS only, no selection), (3) Unguided IPC= 50 $\rightarrow$ herd to IPC= 10 (herding only), and (4) CAGS IPC= 50 $\rightarrow$ herd to IPC= 10 (CAGS + herding). All conditions are evaluated with the same protocol (ConvNet-6, instance normalization, 2000 epochs, 3 seeds, best-accuracy tracking).

Results. Table 24 presents the results. Three findings emerge. First, herding provides a marginal improvement over unguided generation (23.08% vs. 22.79%, +0.29%), confirming that selecting representative images from a larger pool yields a small benefit. Second, CAGS-alone (25.70%) substantially outperforms both herding-alone (23.08%) and the combination of CAGS with herding (24.77%), demonstrating that adaptive guidance at IPC= 10 is more effective than sample selection from a 5$\times$ larger pool. Third, CAGS + herding (24.77%) underperforms CAGS-alone (25.70%) by 0.93%, indicating that herding introduces a slight *redundancy* with CAGS: the diversity signal that herding extracts from the larger pool is already captured by CAGS's per-class guidance adaptation.

Crucially, CAGS improves the quality of the herding pool: the gain from CAGS over unguided is +2.91% without herding and +1.69% with herding (24.77% vs. 23.08%). This confirms that CAGS enhances

Table 24: CAGS vs. herding on ImageNet-100 (100-class, ConvNet-6, 3 seeds). CAGS-alone outperforms both herding-alone and the combination, indicating that adaptive guidance subsumes the benefit of sample selection: the diversity signal that herding extracts from a larger pool is already encoded by CAGS at IPC= 10.

Method	**IPC**	**Top-1 (%)**	**Δ vs. unguided**
Unguided	10	22.79 ± 0.18	—
CAGS (2-factor opt.)	10	25.70 ± 0.07	+2.91
Unguided $\rightarrow$ herding	$50 \rightarrow 10$	23.08 ± 0.07	+0.29
CAGS $\rightarrow$ herding	$50 \rightarrow 10$	24.77 ± 0.29	+1.98

individual sample quality through adaptive guidance—a dimension that sample selection cannot access—while the reduced gain under herding reflects the overlap between the two methods' diversity-improvement mechanisms. In aggregate, the combined method still yields +1.98% over unguided, but the sub-additive structure ($1.98 < 2.91 + 0.29 = 3.20$) confirms partial substitution rather than orthogonality.

These findings have practical implications: CAGS is best used as a standalone generation-time intervention rather than combined with post-hoc sample selection. The result also suggests that CAGS's per-class guidance adaptation and herding's feature-matching selection target overlapping aspects of sample diversity, with CAGS achieving superior results at lower computational cost (single-pass generation at IPC= 10 vs. generating at IPC= 50 then selecting).

6 Limitations

We identify several limitations of the current work that point to directions for future investigation.

Configuration choice is dataset-dependent. The two-factor optimal configuration (α=0, β=0.5, γ=0.5, δ=0) outperforms the four-factor design on most tested datasets (ImageNet-100, ImageNette, ImageWoof, Objects), but the four-factor design wins on certain settings (Animals subset, ResNet-18, IN-200) where mode count and separability exhibit greater variation across classes. This indicates that the optimal complexity metric is not universally fixed but depends on the dataset's class composition. The general framework (Eq. (3)) supports all four factors, and practitioners can tune the weights for their specific dataset using the cached feature pipeline, which makes weight changes nearly free after the one-time VAE extraction. Future work should investigate automatic weight selection based on dataset characteristics.

Exploratory modules not recommended. Two additional modules—TAGS (timestep-adaptive guidance) and IAST (IPC-adaptive guidance range)—were explored but are not recommended. TAGS does not improve over CAGS-alone, and IAST's log-scaling is counterproductive at low IPC. More sophisticated temporal schedules or class-count-conditioned scaling functions are directions for future work.

Evaluation scope and protocol differences. Our experiments use three random seeds per configuration for most settings (three seeds for all ImageNet-1K experiments including ResNet-18 and ResNetAP-10, two seeds for ImageNet-1K ConvNet-6). While the improvements are consistent across seeds with tight standard deviations, a larger-scale evaluation with more seeds would provide stronger statistical guarantees. We evaluate on ImageNet-100 (10-class and 100-class subsets), ImageNette, ImageWoof, semantic subsets (44 and 56 classes), a 200-class ImageNet subset, the full ImageNet-1K (1000 classes) across three architectures, and two non-ImageNet datasets (Food-101, CIFAR-10) with Stable Diffusion as the generator. Our unguided baseline on ImageNette (54.0% with ResNetAP-10) is lower than DMGD's Wang et al. (2026) reported DiT baseline (59.1%), likely due to our heavier augmentation protocol (color jitter, PCA lighting, CutMix with p=1.0) following MGD^3. While our relative improvement over the respective baseline is comparable to DMGD's, the absolute performance gap suggests that evaluation protocol choices significantly impact results, and a unified evaluation protocol across methods would enable fairer comparison. A fair comparison with D4M under the same DiT generator reveals that real-image initialization is sensitive to the noise injection level (t_{start}): a comprehensive sweep over $t_{\text{start}} \in \{5, 10, \ldots, 45\}$ shows that D4M underperforms unguided generation for $t_{\text{start}} \leq 30$ (ranging from 15.06% to 22.58% vs. 22.79% unguided), peaks at t=40 (23.07%), and declines at t=45 (22.42%). This non-monotonic sensitivity highlights a practical advantage of CAGS's principled, hyperparameter-free approach.

The theoretical analysis (Section 4) assumes well-separated Gaussian modes and a second-order approximation of the diversity cost; while the qualitative predictions are confirmed empirically, a more general analysis relaxing these assumptions would strengthen the theoretical contribution.

Generator and domain scope. While we evaluate with two generators—DiT-XL/2 on ImageNet-derived datasets and Stable Diffusion 1.5 on Food-101 and CIFAR-10—the non-ImageNet results use SD 1.5, which was not specifically trained for the target domains, resulting in low absolute accuracy on Food-101 (4.14%). Domain-specific generators (e.g., food-image diffusion models) would likely yield higher absolute performance and stronger evidence for the adaptive guidance principle. Additionally, the IPC=1 result for the optimal two-factor configuration on the 100-class subset is not yet available under the consistent 2000-epoch protocol, leaving a minor gap in the IPC ablation. Future work should evaluate with domain-adapted generators and complete the IPC sweep.

7 Conclusion

We have presented CAGS (Class-Adaptive Guidance Strength), a training-free method that replaces the globally fixed classifier-free guidance of MGD^3 with per-class adaptive guidance. CAGS assigns per-class guidance strengths based on a complexity metric combining cluster entropy and intra-class variance—the two factors identified as effective through systematic factor ablation and weight grid search from an initial set of four candidate factors. Mode count and inter-class separability are eliminated: the former is nearly constant across classes (K=2 for 94% of classes), and the latter produces near-constant guidance (std 0.006). We provide a rigorous theoretical explanation (Section 4): under a Gaussian mixture model, the optimal per-class guidance $\lambda_c^* \propto v_c/\sigma_c^2$ is increasing in both intra-class variance v_c and cluster entropy H_c (since $\sigma_c^2 = \text{Var}[\log \pi_k]$ is decreasing in H_c), and is independent of inter-class separability. An adaptivity gap theorem (Theorem 1) proves that the gain from per-class adaptation over fixed guidance is proportional to the cross-class variance of λ_c^*, which increases with class count—providing a theoretical foundation for the empirical scaling result. The optimal two-factor configuration (β=0.5, γ=0.5) achieves $25.70 \pm 0.07\%$ on ImageNet-100, surpassing the four-factor default ($25.29 \pm 0.27\%$) with 3.9× lower variance. Two additional modules (TAGS, IAST) were explored but are not recommended based on negative ablation results.

Our experiments reveal three key insights. First, *CAGS provides substantial and consistent gains when class complexity variation is large*: on the 100-class subset, the optimal CAGS improves by +12.8% relative at IPC=10 over unguided, while fixed guidance *hurts* (−6.5%). The gains scale on deeper architectures: +15.1%–17.6% on ResNet-18 and +10.8%–15.2% on ResNetAP-10, both with ranges exceeding the +12.8%

gain on ConvNet-6, confirming that the improvement stems from richer discriminative features in the synthetic data. On the 10-class subset, CAGS provides a large gain at IPC=50 (+7.9%) and generalizes to ImageNette (+10.6%). Crucially, scaling to 200 classes amplifies the CAGS advantage to +3.30% absolute (+17.8% relative), and scaling to the full ImageNet-1K (1000 classes) further amplifies it to +3.58% absolute (+31.2% relative) on ConvNet-6, +6.97% absolute (+41.3% relative) on ResNet-18, and +6.83% absolute (+43.7% relative) on ResNetAP-10, confirming that per-class adaptation becomes increasingly beneficial as the complexity distribution widens. A fair comparison with D4M under the same DiT generator reveals that real-image initialization is highly sensitive to the denoising start timestep: across a sweep of $t_{\text{start}} \in \{5, 10, \ldots, 45\}$, D4M accuracy rises steadily from 15.06% at t=5 to a peak of 23.07% at t=40, then declines to 22.42% at t=45. At t_{start}=25, it substantially underperforms unguided generation (20.21% vs. 22.79%), while even at the optimal t=40, it falls 2.63% short of CAGS (23.07% vs. 25.70%) and requires real-image access and an additional hyperparameter. This confirms that CAGS's training-free, hyperparameter-free approach—generating from pure noise with adaptive per-class guidance—is more principled and effective than real-image initialization.

Second, *systematic simplification improves both accuracy and stability on most datasets*: the weight grid search over 12 configurations reveals that eliminating mode count ($\alpha \to 0$) and separability ($\delta \to 0$) while balancing entropy and intra-class variance ($\beta \approx \gamma$) yields the best performance on ImageNet-100, ImageNette, ImageWoof, and Objects. The optimal two-factor design amplifies the benefit on fine-grained datasets where the four-factor configuration provides only marginal gains (ImageWoof: +0.8% $\to$ +3.5% relative). However, the four-factor design retains an advantage on certain class compositions (Animals subset, ResNet-18, IN-200), indicating that mode count and separability capture complementary signals when these properties vary substantially across classes. This nuanced finding suggests that the optimal complexity metric is dataset-dependent, and the two-factor configuration's robustness and lower variance make it the recommended default.

Third, *per-class adaptation is the key mechanism*: our factor ablation reveals that cluster entropy is the most effective single complexity factor, while inter-class separability—which produces near-constant guidance—is the least effective. Our theoretical analysis (Section 4) explains this: the optimal per-class guidance $\lambda_c^* \propto v_c/\sigma_c^2$ is increasing in entropy H_c (Proposition 2), while separability—an inter-class property—does not affect the intra-class optimal guidance (Lemma 2). We further validate generalization beyond ImageNet: CAGS yields a +82.4% relative improvement on Food-101 (101 classes) using Stable Diffusion 1.5, while remaining neutral on CIFAR-10 (10 classes, −0.4%), confirming that the scaling boundary condition holds across generators and domains.

CAGS occupies a unique position in the landscape of diffusion-based dataset distillation: unlike LGD Chan-Santiago and Shah (2026), which requires multiple training rounds for iterative sample selection at higher IPC budgets, and unlike DMGD Wang et al. (2026), which requires optimal transport computation for distribution matching, CAGS preserves the training-free, single-pass paradigm of MGD^3 while delivering consistent improvements through principled guidance adaptation with negligible computational overhead. A complementarity analysis (Section 5.21) reveals that CAGS subsumes the benefit of post-hoc sample selection: CAGS-alone (25.70%) outperforms both herding-alone (23.08%) and the combination of CAGS with herding (24.77%), indicating that adaptive guidance already encodes the diversity signal that sample selection would extract from a larger pool. Future directions include extending the framework to domain-specific diffusion models, investigating automatic weight selection, and evaluating whether the entropy-driven findings generalize to other diffusion architectures.

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